Nonlinear scalar scattering on Kerr: resonant response, frequency tails, and principal-mode enhancement

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We investigate the first-order scalar energy-flux response to local interactions $|\Phi|^{2n}\Phi$ on a fixed Kerr background across the infrared, resonance, and post-resonance frequency regimes. For cubic axisymmetric scattering at $a/M = 0.5$, $\ell = 0, 1, 2$, the nonlinear absorption has Breit–Wigner-type peaks near the fundamental scalar quasinormal-mode frequency scale. Post-resonance tails favor an exponential over a power law in the tested windows. An exact Green-function decomposition separates flux and phase corrections. Algebraic-prefactor tests quantify the dependence of the fitted decay scale on the tail model. At fixed incident amplitude and subextremal spin, cubic axisymmetric absorption and its first-order correction have the same $(M\omega)^{2\ell+2}$ power, with a spin-independent limiting ratio. For the monopole this ratio is $\ln 2/\pi$; higher interaction powers give $M^{2n-2}\chi^{(n)}_{H,00} \sim c_n(M\omega)^{2n-2}$. Near extremality at fixed p/κ, principal modes with $h = 1/2 + i\beta$ concentrate the incident-normalized field in the throat. Matching hypergeometric and Whittaker solutions and evaluating the nonlinear Green integral give $\widehat{\chi}_n = \sigma^{-n}\mathcal{F}_n(p/\kappa, \beta\ln\sigma) + o(\sigma^{-n})$, where $\sigma = (r_+ - r_-)/M$. The coefficient has logarithmic period π/β and is specified with zero fitted parameters. Spectral calculations test $(\ell, m) = (2,2), (3,3), (4,4)$, $n = 1, 2, 3$, and both signs of the horizon detuning. Independent spectral and quadrature refinements, coefficient tails, field-dependent radial residuals and flux interference checks quantify numerical accuracy. The enhancement identifies a shrinking weak-coupling domain near extremality. Flux-response zeros coexist with enhanced phase coefficients at selected near-critical detunings. All results use fixed asymptotic incident normalization.

I. INTRODUCTION

Kerr wave propagation and a local field interaction jointly determine the magnitude and phase of nonlinear scattering. We quantify their combined effect through the complex first-order response at fixed incident radiation. Its interference with the linear field determines the energy exchanged between the horizon and infinity; its imaginary component determines the transmitted phase shift. This common observable organizes the finite-frequency profile and the two asymptotic limits studied below.

Black-hole scattering relates incident radiation to the flux reaching the future horizon and the flux returning to infinity. Linear scalar scattering in Schwarzschild and Kerr spacetimes is well established [1–3]. Its absorption and reflection coefficients provide real-frequency observables complementary to quasinormal-mode (QNM) spectra[4–6]. The nonlinear response resolves the frequency dependence of absorption through resonance peaks and a decaying post-resonance tail. At the low-frequency endpoint, linear axisymmetric absorption vanishes and its fractional nonlinear response can remain finite. Near extremality, incident-normalized principal modes develop enhanced throat amplitudes. These effects depend on the scattering Green function, the nonlinear source, and the normalization of the driving field. The first-order energy-flux response follows from interference between the linear and sourced fields at fixed incident radiation.

Several nonlinear problems must be distinguished. Numerical evolutions of incident gravitational radiation measure waveforms and Bondi fluxes [7, 8]; self-interacting massive scalar clouds instead describe bound-state growth, level transfer, and collapse [9, 10]. Nonlinear Q-clouds are synchronized bound states with decay at infinity[11]. Here the field is massless, the driving frequency is real, and radiation is supplied from infinity. The calculation treats a stationary driven response on a prescribed background. Cloud evolution studies bound-state dynamics and spacetime evolution includes gravitational backreaction. Gravitational self-force theory adds a worldline source, regularization, and gauge control [12, 13].

Recent stationary gravitational scattering calculations separate the homogeneous scattering kernel from a source-dependent nonlinear susceptibility[14]. Near-extremal linear response and critical scaling have also been studied for finite initial perturbations[15, 16], while weakly nonlinear gravitational QNM interactions admit a distinct slow-time dynamical description[17]. The present calculation fixes a monochromatic incident amplitude and measures derivatives of scattering fluxes and transmitted phase with respect to a local scalar coupling. Its illumination and normalization conditions specify the relation to these dynamical problems.

Building on the fixed-background Green-function construction for Schwarzschild scattering[18], we determine the frequency-dependent nonlinear scalar scattering on

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Kerr. The resonance and tail analysis follows the logic of the Schwarzschild reference: construct the sourced field, read its interference contribution from the energy flux, and compare the resulting frequency profile with the QNM and horizon scales. The asymptotic analysis then determines how the same response changes when the incident wavelength becomes large or the Kerr throat becomes long. These two limits link the interaction power to infrared suppression and principal-mode enhancement, respectively. Explicit connection coefficients and Green projections determine the response under a common fixed-incident normalization. The Kerr formulation requires three changes. The horizon frequency is $p = \omega - m\Omega_H$, angular functions are spheroidal, and the covariant source contains the factor $\Sigma = r^2 + a^2\cos^2\theta$. These changes determine the radial operator, projected source and incident-normalized flux response. The exact relation $T^{(1)} + R^{(1)} = 0$ then constrains the first-order flux response, while off-diagonal angular channels first contribute to the integrated energy flux at second order in the coupling.

Our main results separate the effects of black-hole penetration and local self-interaction. The fixed-spin cubic sweep exhibits resonance peaks with stable fitted centers and an exponentially decaying post-resonance response. The exact Green-function decomposition identifies the part that changes the absorbed flux and the part that shifts the horizon phase. The peak centers, tail slopes, prefactor tests, and their numerical error budgets are reported separately. For cubic axisymmetric self channels, matching the regular static solution to the far-zone wave gives the same $\omega^{2\ell+2}$ power for linear absorption and its first-order correction. The spin-dependent penetration factor cancels in their ratio, whose leading coefficient is fixed by a far-zone Bessel integral. The monopole admits the explicit complex response $\rho_{H,00} \to \ln 2/(2\pi) + i/4$: the absorbed-flux correction and the horizon-phase correction are distinct parts of the same response. For $|\Phi|^{2n}\Phi$ the monopole fractional absorption response scales as ω^{2n-2} with an analytically computable positive coefficient. The cubic case therefore remains unsuppressed relative to linear absorption, whereas every higher local power is infrared suppressed. This hierarchy classifies the specified fixed-incident scattering observable.

The fixed-spin infrared limit isolates the source response after the penetration factor cancels in the fractional observable. To examine a geometric enhancement of the driving field, the second asymptotic path holds $y = p/\kappa$ fixed while $a/M \to 1$. For principal modes, $h = 1/2 + i\beta$ fixes the throat density envelope to κ^{-1} and the nonlinear source produces a κ^{-n} relative response envelope. Two-branch interference supplies a log-periodic coefficient. We derive that coefficient by combining the analytic near-superradiant connection[19] with the covariant nonlinear Green projection. A convergent throat integral and matching denominators bounded away from zero specify the domain of this expansion. Real-weight modes require the corresponding upper-limit convergence analysis and full outer matching.

The fixed-spin frequency sweep uses $a/M = 0.5$, $m = 0$, and $\ell = 0, 1, 2$. The new principal-response grid contains 81 physical parameter points: three principal modes, three interaction orders, three nonzero detunings, and three near-extremal spins. Independent spectral and quadrature refinements test the nonlinear coefficients. The two sets of experiments use the same covariant source and incident normalization.

Section II defines the field equation, source projection, and flux observables. Section III describes the endpoint spectral solver, the Green-function interpretation of the frequency profile, and the independent accuracy diagnostics. Section IV first presents the resonance peaks and high-frequency tail tests, then derives and verifies the infrared and principal-mode responses and examines the Schwarzschild limit. The final section discusses the physical interpretation and remaining limitations. We use $G = c = \hbar = 1$, metric signature $(-,+,+,+)$, and the Fourier convention $e^{-i\omega t + im\phi}$. Numerical frequencies are given in units $M = 1$ unless stated otherwise.

II. NONLINEAR SCATTERING PROBLEM IN KERR SPACETIME

In Boyer-Lindquist coordinates the Kerr metric is

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1 - \frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\mathrm{d}t\,\mathrm{d}\phi \\ &+ \frac{\Sigma}{\Delta}\mathrm{d}r^2 + \Sigma\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\mathrm{d}\phi^2, \end{aligned} \tag{1}$$

where

$$\Sigma = r^2 + a^2\cos^2\theta, \quad \Delta = (r - r_+)(r - r_-), \tag{2}$$

$$A = (r^2 + a^2)^2 - a^2\Delta\sin^2\theta. \tag{3}$$

The horizon radii and angular velocity are

$$r_\pm = M \pm \sqrt{M^2 - a^2}, \qquad \Omega_H = \frac{a}{r_+^2 + a^2}. \tag{4}$$

We consider a weak cubic self-interaction

$$\Box\Phi + \epsilon|\Phi|^2\Phi = 0, \qquad \epsilon \in \mathbb{R}, \quad |\epsilon| \ll 1, \tag{5}$$

Here ϵ is a signed perturbative coefficient. With the action convention in Eq. (7), a conventional positive quartic potential $V = \lambda|\Phi|^4/2$ corresponds to $\epsilon = -\lambda$; therefore the sign statements below always refer to Eq. (5), not to a separate stability assumption about the scalar potential. In four spacetime dimensions the canonical scalar has dimensions $[\Phi] = M^{-1}$, so ϵ is dimensionless in the action convention below. For a field amplitude that is not absorbed into the unit-horizon normalization, a useful local control parameter is

$$\chi_{\mathrm{nl}} = |\epsilon| M^2 \max_{\mathcal{D}} |\Phi^{(0)}|^2 \ll 1, \tag{6}$$

where $\mathcal{D}$ is the radial and angular region that dominates the cubic source projection. The reported $T^{(1)}$ and $R^{(1)}$ are coefficients in the derivative of the fixed-incident flux with respect to ϵ at $\epsilon = 0$; a finite-coupling application must additionally check that $|\epsilon||\mathcal{A}_{\rm in}|^2|T^{(1)}| \ll |T^{(0)}|$ and the analogous reflected-flux condition. We do not interpret the present frequency-domain calculation as a test of nonlinear saturation or of the expansion beyond this perturbative regime. This equation follows from the fixed-background scalar action

$$S_\Phi = \int \mathrm{d}^4x\,\sqrt{-g}\left[-g^{\mu\nu}\nabla_\mu\Phi^*\nabla_\nu\Phi + \frac{\epsilon}{2}|\Phi|^4\right]. \tag{7}$$

The stress tensor used below is the Hilbert stress tensor of the same effective field theory,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_\Phi}{\delta g^{\mu\nu}}. \tag{8}$$

For later use, the explicit tensor and the associated stationary Killing energy flux are

$$T_{\mu\nu} = \nabla_\mu\Phi^*\nabla_\nu\Phi \tag{9}$$

$$+ \nabla_\nu\Phi^*\nabla_\mu\Phi + g_{\mu\nu}\left[-\nabla_\alpha\Phi^*\nabla^\alpha\Phi + \frac{\epsilon}{2}|\Phi|^4\right], \tag{10}$$

$$j^\mu_E = -T^\mu{}_\nu\xi^\nu = -T^\mu{}_t, \qquad \xi = \partial_t,$$

$$\mathcal{F}^{\rm out}_E(r) = \int_0^{2\pi}\mathrm{d}\phi\int_0^\pi \mathrm{d}\theta\,\sqrt{-g}\,j^r_E, \tag{11}$$

$$\mathcal{F}^H_E = -\lim_{r\to r_+}\mathcal{F}^{\rm out}_E(r). \tag{12}$$

The reversed horizon orientation in the last definition makes positive $\mathcal{F}^H_E$ correspond to absorption when $p > 0$. The quartic term is proportional to $\delta^r{}_t$ in $T^r{}_t$ and therefore does not itself carry radial flux; the nonlinear boundary correction comes from the interference of the zeroth- and first-order kinetic terms. We now expand the field as

$$\Phi = \Phi^{(0)} + \epsilon\Phi^{(1)} + \mathcal{O}(\epsilon^2). \tag{13}$$

The zeroth-order incident wave is a single scalar spheroidal mode,

$$\Phi^{(0)} = e^{-\mathrm{i}\omega t + \mathrm{i}m\phi}S_{\ell m}(\theta; a\omega)R_{\ell m\omega}(r), \tag{14}$$

with angular normalization

$$\int |S_{\ell m}|^2\mathrm{d}\Omega = 1. \tag{15}$$

The scalar spheroidal harmonic obeys

$$\left[\frac{1}{\sin\theta}\partial_\theta\sin\theta\partial_\theta + a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta} + A_{\ell m}\right]S_{\ell m} = 0. \tag{16}$$

The radial Teukolsky equation for $s = 0$ is [20]

$$\mathcal{L}_{\ell m\omega}R \equiv \frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \left(\frac{K^2}{\Delta} - \lambda_{\ell m\omega}\right)R = 0, \tag{17}$$

where

$$K = (r^2+a^2)\omega - am, \qquad \lambda_{\ell m\omega} = A_{\ell m} + a^2\omega^2 - 2am\omega. \tag{18}$$

The endpoint normalization used here can be related to the analytic MST solutions, which provide independent low-frequency connection formulae [21]. The numerical results below are assessed using internal spectral convergence and flux-balance diagnostics. The tortoise coordinate is defined by

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{r^2+a^2}{\Delta}, \tag{19}$$

and the horizon wave number is

$$p = \omega - m\Omega_H. \tag{20}$$

The ingoing solution is normalized to unit amplitude at the future horizon,

$$R^{\rm in}_{\ell m\omega} \sim e^{-\mathrm{i}pr_*}, \qquad r\to r_+, \tag{21}$$

and at infinity it has the form

$$R^{\rm in}_{\ell m\omega} \sim B^{\rm inc}\frac{e^{-\mathrm{i}\omega r_*}}{r} + B^{\rm ref}\frac{e^{+\mathrm{i}\omega r_*}}{r}, \qquad r\to\infty. \tag{22}$$

With this convention the linear flux coefficients are

$$T^{(0)} = \frac{p(r_+^2+a^2)}{\omega|B^{\rm inc}|^2}, \qquad R^{(0)} = \frac{|B^{\rm ref}|^2}{|B^{\rm inc}|^2}. \tag{23}$$

For clarity, these coefficients are energy-flux ratios rather than merely amplitude ratios. Let $\xi = \partial_t$ be the stationary Killing field and let $T_{\mu\nu}$ be the scalar stress tensor defined in Eq. (8). The outward-oriented Killing energy current is $j^\mu_E = -T^\mu{}_t$, so at a surface of constant r we use $\mathcal{F}^{\rm out}_E = \int j^r_E\sqrt{-g}\,\mathrm{d}\theta\mathrm{d}\phi$. When quoting absorption by the future horizon, the horizon boundary orientation is reversed, so that positive absorption corresponds to positive $\mathcal{F}^H_E = -\mathcal{F}^{\rm out}_E|_{r=r_+}$ for $p > 0$. For a single Fourier mode, the energy current is the mode frequency times the radial Klein–Gordon current, up to one common positive normalization $\mathcal{N}$. Consequently,

$$\begin{aligned}\mathcal{F}^{\rm in}_{E,\infty} &= \mathcal{N}\omega^2|B^{\rm inc}|^2,\\ \mathcal{F}^{\rm ref}_{E,\infty} &= \mathcal{N}\omega^2|B^{\rm ref}|^2,\\ \mathcal{F}_{E,H} &= \mathcal{N}\omega p(r_+^2+a^2).\end{aligned} \tag{24}$$

Their ratios give Eq. (23), including the sign of the horizon flux when $p < 0$. The self-interaction contributes a term proportional to $\delta^r{}_t$ in the mixed stress tensor and therefore does not add a radial boundary flux; at first order, the nonlinear flux correction is the interference term between $\Phi^{(0)}$ and $\Phi^{(1)}$ derived below.

For real ω the conserved radial current gives

$$T^{(0)} + R^{(0)} = 1. \tag{25}$$

At first order,

$$\Box\Phi^{(1)} = -|\Phi^{(0)}|^2\Phi^{(0)}. \tag{26}$$

Because the scalar is complex and the interaction is $U(1)$ invariant, $|e^{-i\omega t+im\phi}|^2 e^{-i\omega t+im\phi}$ has the same (ω, m) dependence as the incident mode. We therefore expand

$$\Phi^{(1)} = e^{-\mathrm{i}\omega t+\mathrm{i}m\phi}\sum_{\ell'\geq|m|} S_{\ell' m}(\theta; a\omega)R^{(1)}_{\ell'|\ell m\omega}(r). \tag{27}$$

After multiplying by Σ and projecting onto a target spheroidal harmonic, the radial source is

$$\mathcal{L}_{\ell' m\omega}R^{(1)}_{\ell'|\ell m\omega} = -Q_{\ell'\ell m}(r), \tag{28}$$

where

$$Q_{\ell'\ell m}(r) = \left[r^2 C^{(0)}_{\ell'\ell m} + a^2 C^{(2)}_{\ell'\ell m}\right]|R^{\rm in}_{\ell m\omega}|^2 R^{\rm in}_{\ell m\omega}, \tag{29}$$

and

$$C^{(j)}_{\ell'\ell m} = \int \mathrm{d}\Omega\, \cos^j\theta\, S^*_{\ell' m}|S_{\ell m}|^2 S_{\ell m}, \qquad j = 0, 2. \tag{30}$$

The self channel $\ell' = \ell$ is the only target channel that interferes with the linear outgoing wave in the total energy flux at order ϵ.

III. METHODS

A. Pseudospectral method for homogeneous solutions

The homogeneous radial equation is solved in the compact coordinate

$$z = \frac{r_+}{r}, \qquad z = 0 \text{ at infinity}, \qquad z = 1 \text{ at the horizon}. \tag{31}$$

The oscillatory endpoint phases are peeled off before Chebyshev collocation. This endpoint-first organization is also consistent with independent spectral black-hole perturbation work, where the asymptotic behavior is factored before the Chebyshev representation is truncated [22]:

$$R_{\rm in} = e^{-\mathrm{i}pr_*}u_{\rm in}(z), \tag{32}$$

$$R_{\rm down} = \frac{e^{-\mathrm{i}\omega r_*}}{r}u_{\rm down}(z), \tag{33}$$

$$R_{\rm up} = \frac{e^{+\mathrm{i}\omega r_*}}{r}u_{\rm up}(z). \tag{34}$$

The functions $u_{\rm in}$, $u_{\rm down}$, and $u_{\rm up}$ are regular on their respective subdomains. The degenerate endpoint rows of the transformed ODE therefore supply the boundary regularity conditions. The outer and inner domains are matched at a finite radius $r_{\rm m}$ by continuity of R and $\mathrm{d}R/\mathrm{d}r$.

More explicitly, after the phase peeling the collocation equation has the form

$$B_2(z)u_{,zz} + B_1(z)u_{,z} + B_0(z)u = 0. \tag{35}$$

These coefficients follow directly without a further field redefinition. Writing $R = f(r)u[z(r)]$, $g = \mathrm{d}z/\mathrm{d}r = -z^2/r_+$, and $V = K^2/\Delta - \lambda_{\ell m\omega}$, one obtains

$$\begin{aligned} B_2 &= \Delta g^2, \\ B_1 &= \Delta\left(g_{,r} + 2g\frac{f_{,r}}{f}\right) + \Delta_{,r}g, \\ B_0 &= \Delta\frac{f_{,rr}}{f} + \Delta_{,r}\frac{f_{,r}}{f} + V, \end{aligned} \tag{36}$$

where $f = e^{-ipr_*}$ for the inner ingoing branch and $f = r^{-1}e^{\mp i\omega r_*}$ for the outer down/up branches. The expressions used in the code are the analytic endpoint limits of Eq. (36), so no finite-difference evaluation of a singular quotient is required.

For the outer branch $f = r^{-1}e^{i\sigma\omega r_*}$, with $\sigma = -1$ for down and $+1$ for up, the exact limits are

$$B_2(0) = 0, \quad B_1(0) = -2i\sigma r_+\omega, \quad B_0(0) = -(\lambda + 2am\omega). \tag{37}$$

The last expression is $-(A + a^2\omega^2)$ with $\lambda = A + a^2\omega^2 - 2am\omega$. It reduces to $-\lambda$ for the axisymmetric numerical samples. The exact endpoint row is checked directly against the transformed differential equation for all m.

At the exact endpoint of each branch B_2 vanishes, so the endpoint row is the endpoint compatibility relation $B_1u_{,z} + B_0u = 0$. The neighboring row is replaced by the normalization $u_{\rm down}(0) = u_{\rm up}(0) = 1$ on the outer domain or $u_{\rm in}(1) = 1$ on the inner domain. Thus the horizon is included as an exact collocation endpoint. For nonzero frequency, infinity is an irregular singular point of the compactified outer equation. Phase peeling removes the leading oscillation; endpoint regularity of the remaining amplitude is tested through its spectral convergence. Spectral coefficient decay and mesh-refinement tests therefore remain essential [18]. Each domain uses $N+1$ Chebyshev points and a dense complex linear solve. For $|M\omega| < 0.1$ we use the mapped nodes

$$z(y) = z_L + (z_R - z_L)\frac{\sinh[\kappa(1-y)/2]}{\sinh\kappa}, \qquad -1 \leq y \leq 1, \tag{38}$$

with $\kappa_{\rm out} = |\log(2M|\omega|)|$ and $\kappa_{\rm in} = \kappa_{\rm out}/2$; otherwise the map is linear. This redistributes the same endpoint-inclusive Chebyshev grid toward the outer end of each subdomain and does not move either physical boundary. The low-frequency and resonance-region production runs use $N = 240$ on both domains, $r_{\rm m} = 40M$, Gauss order $q = 192$ for the compact inner integral, and relative tolerance 10^{-11} for the oscillatory phase-channel tail. The high-frequency fits use 43 points in the three tail-fit windows at the same N and $r_{\rm m}$ but with $q = 2048$. The larger order is required because the compact horizon-side Green integral remains oscillatory after the homogeneous

endpoint phases have been peeled. The tail-order audit compares $q = 1792$ with $q = 2048$ and independently lowers the radial order to $N = 224$. The low-frequency runs use the automatic map in Eq. (38); all reported results are obtained in IEEE double precision and are checked by the order, matching-radius, quadrature, Wronskian, and flux scans reported below.

The angular eigenvalue is computed by diagonalizing the truncated spherical harmonic representation of $\ell(\ell+1) - (a\omega)^2 \cos^2\theta$, retaining sixteen modes above the requested ℓ. The source projectors are evaluated with normalized spheroidal harmonics and $q_\theta = 256$ point Gauss–Legendre quadrature. The separate basis and quadrature audit in Table I compares these active settings with 24 extra modes and $q_\theta = 512$. For the radial Green integrals, the compact inner part is integrated in z with the Jacobian $\mathrm{d}r = r_+/z^2\,\mathrm{d}z$, while the outer part is decomposed into its finite phase channels in r_* before oscillatory quadrature.

The direct Teukolsky potential is long ranged, so an alternative numerical route is to transform the equation to a short-ranged generalized Sasaki–Nakamura equation [23, 24]. We retain the direct Teukolsky variable here because it keeps the covariant Σ-weighted source and the endpoint normalization in the same representation. The GSN formulation is consequently a useful independent route for future high- frequency or extended-source checks, but it is not used to define any of the production values reported below.

The scattering relation used in the code is the usual in/down/up decomposition,

$$R_{\rm in} = B^{\rm inc} R_{\rm down} + B^{\rm ref} R_{\rm up}. \tag{39}$$

At $r = r_{\rm m}$ the amplitudes are obtained from

$$\begin{pmatrix} R_{\rm down} & R_{\rm up} \\ R'_{\rm down} & R'_{\rm up} \end{pmatrix}_{r_{\rm m}} \begin{pmatrix} B^{\rm inc} \\ B^{\rm ref} \end{pmatrix} = \begin{pmatrix} R_{\rm in} \\ R'_{\rm in} \end{pmatrix}_{r_{\rm m}}. \tag{40}$$

Solving this two-by-two matching system gives $B^{\rm inc}$ and $B^{\rm ref}$ and therefore the linear coefficients in Eq. (23). The residual $|T^{(0)} + R^{(0)} - 1|$ is used as the first check of the homogeneous solve.

B. Nonlinear scattering coefficients

1. *The nonlinear balance law*

Let $R^{(\ell)}_{\rm up}$ denote the target-channel homogeneous solution that is purely outgoing at infinity. The Green-function solution in the self channel has outgoing amplitude at infinity and ingoing amplitude at the horizon. The conserved Wronskian is

$$W_\ell = \Delta\left(R^{\rm in}_{\ell m\omega}\frac{\mathrm{d}R^{\rm up}_{\ell m\omega}}{\mathrm{d}r} - R^{\rm up}_{\ell m\omega}\frac{\mathrm{d}R^{\rm in}_{\ell m\omega}}{\mathrm{d}r}\right), \tag{41}$$

which is independent of r for two homogeneous solutions of Eq. (17). The Green-function amplitudes are

$$A^{(1)}_{\rm ref} = -\frac{1}{W_\ell}\int_{r_+}^{\infty} Q_{\ell\ell m}(r) R^{\rm in}_{\ell m\omega}(r)\,\mathrm{d}r, \tag{42}$$

$$A^{(1)}_{H} = -\frac{1}{W_\ell}\int_{r_+}^{\infty} Q_{\ell\ell m}(r) R^{\rm up}_{\ell m\omega}(r)\,\mathrm{d}r, \tag{43}$$

where the minus sign follows from $\mathcal{L}_{\ell m\omega} R^{(1)} = -Q_{\ell\ell m}$. In the numerical implementation the Wronskian is evaluated from the matched homogeneous solutions and checked against its constancy. The amplitudes $A^{(1)}_{\rm ref}$ and $A^{(1)}_{H}$ are quoted in the unit-horizon normalization of Eq. (21). To form scattering coefficients one must instead hold the incident wave fixed. If the zeroth-order field is rescaled by $\alpha = 1/B^{\rm inc}$, the cubic source scales as $|\alpha|^2\alpha$. Therefore the fixed-incident first-order boundary amplitudes are

$$A^{(1)}_{\rm ref,fix} = \frac{A^{(1)}_{\rm ref}}{|B^{\rm inc}|^2 B^{\rm inc}}, \qquad A^{(1)}_{H,\rm fix} = \frac{A^{(1)}_{H}}{|B^{\rm inc}|^2 B^{\rm inc}}. \tag{44}$$

The zeroth-order fixed-incident horizon amplitude is $1/B^{\rm inc}$. It is useful to retain the complex relative correction

$$\rho_H(\omega) \equiv \frac{A^{(1)}_{H,\rm fix}}{1/B^{\rm inc}} = \frac{A^{(1)}_{H}}{|B^{\rm inc}|^2}. \tag{45}$$

For unit incident amplitude the total horizon coefficient is therefore $(1/B^{\rm inc})[1 + \epsilon\rho_H + \mathcal{O}(\epsilon^2)]$. To first order, $\mathrm{Re}\,\rho_H$ changes its modulus whereas $\mathrm{Im}\,\rho_H$ shifts its phase. Only the former enters an energy flux at this order. Expanding the horizon and infinity fluxes of the fixed-incident field then gives

$$T^{(1)} = \frac{2p(r_+^2 + a^2)}{\omega|B^{\rm inc}|^4}\,\mathrm{Re}\,A^{(1)}_{H}, \tag{46}$$

$$R^{(1)} = \frac{2\,\mathrm{Re}(\overline{B^{\rm ref}}\,A^{(1)}_{\rm ref})}{|B^{\rm inc}|^4}. \tag{47}$$

For $p \neq 0$, division by the linear horizon coefficient removes the explicit superradiant factor and defines the threshold-regular relative response

$$\chi_H(\omega) \equiv \frac{T^{(1)}}{T^{(0)}} = \frac{2\,\mathrm{Re}\,A^{(1)}_{H}}{|B^{\rm inc}|^2} = 2\,\mathrm{Re}\,\rho_H. \tag{48}$$

The right-hand side remains directly evaluable as $p \to 0$. Provided the real-frequency homogeneous amplitudes and the projected Green integral are smooth there,

$$\begin{aligned} T^{(0)} &= p\tau_0 + \mathcal{O}(p^2), \\ T^{(1)} &= p\tau_1 + \mathcal{O}(p^2), \\ \chi_H^\star &= \lim_{p\to 0}\chi_H = \frac{\tau_1}{\tau_0}. \end{aligned} \tag{49}$$

This form avoids a numerical 0/0 operation and separates the physical vanishing of the horizon flux from cancellation in the reflected interference term. In the superradiant regime the amplification factor $Z \equiv R - 1 = -T$ consequently obeys

$$Z = Z^{(0)}\left[1 + \epsilon|\mathcal{A}_{\rm in}|^2\chi_H\right] + \mathcal{O}(\epsilon^2|\mathcal{A}_{\rm in}|^4), \qquad p < 0. \quad (50)$$

Equivalently, with the incident energy flux held fixed,

$$T^{(1)} = \frac{1}{\mathcal{F}^{\rm in}_{E,\infty}} \frac{\partial \mathcal{F}^{H}_{E}}{\partial \epsilon}\bigg|_{\epsilon=0}, \qquad R^{(1)} = \frac{1}{\mathcal{F}^{\rm in}_{E,\infty}} \frac{\partial \mathcal{F}^{\rm ref}_{E,\infty}}{\partial \epsilon}\bigg|_{\epsilon=0}. \quad (51)$$

Thus the quantities plotted below are normalized changes of the horizon and infinity energy fluxes, rather than changes of radial-wave amplitudes alone. They are quoted for unit incident amplitude. For a general dimensionless incident amplitude $\mathcal{A}_{\rm in}$, homogeneity of the cubic source gives

$$\begin{aligned} T &= T^{(0)} + \epsilon|\mathcal{A}_{\rm in}|^2 T^{(1)} + \mathcal{O}(\epsilon^2|\mathcal{A}_{\rm in}|^4), \\ R &= R^{(0)} + \epsilon|\mathcal{A}_{\rm in}|^2 R^{(1)} + \mathcal{O}(\epsilon^2|\mathcal{A}_{\rm in}|^4). \end{aligned} \quad (52)$$

Accordingly, $T^{(1)}$ and $R^{(1)}$ are response coefficients per unit incident-amplitude squared; unlike their linear counterparts, they are not independent of the incident intensity. The current conservation identity gives the first-order balance law

$$T^{(1)} + R^{(1)} = 0. \quad (53)$$

The local reason for this identity is visible directly in the radial equation. Let $J^{(1)}$ be the term linear in ϵ in the radial Klein–Gordon current. The cross-current identity gives

$$\frac{{\rm d}J^{(1)}}{{\rm d}r} = -\,{\rm Im}\Big[\left(R^{\rm in}_{\ell m\omega}\right)^{*} Q_{\ell\ell m}(r)\Big] = 0, \quad (54)$$

because the self-channel projectors are real for a real spheroidal parameter and $Q_{\ell\ell m} \propto |R^{\rm in}|^2 R^{\rm in}$. Thus the smooth cubic source changes the two boundary currents through interference, but does not inject net Killing energy into the stationary mode at this perturbative order. Equation (53) is the boundary form of this local identity, while the field residual and spectral tail remain independent discretization diagnostics. Numerically this equation is the most useful global precision diagnostic, because it tests the homogeneous normalization, the nonlinear Green integral, and the flux normalization simultaneously. The fixed-incident normalization and the appearance of the $|B^{\rm inc}|^4$ denominators are derived explicitly in Appendix A.

2. *Numerical accuracy diagnostics*

The accuracy diagnostics used below are internal to the spectral calculation. Let the homogeneous equation be written at a collocation point as

$$\mathcal{R}_R = \Delta R'' + \Delta' R' + \left(\frac{K^2}{\Delta} - \lambda_{\ell m\omega}\right) R. \quad (55)$$

The reported relative residual is

$$\delta_R = \frac{|\mathcal{R}_R|}{|\Delta R''| + |\Delta' R'| + |(K^2/\Delta - \lambda_{\ell m\omega})\, R|}. \quad (56)$$

Thus the denominator contains the actual local field amplitude, not only the differential-equation coefficients. This makes the residual insensitive to an overall rescaling of the homogeneous solution. The original radial residual is evaluated independently on 600 Chebyshev–Gauss points per subdomain that are not used in the collocation solve. The phase-peeled Chebyshev series is differentiated analytically at these off-grid points before reconstructing the three terms in Eq. (56); the degenerate endpoint rows remain reserved for the boundary regularity conditions. We also monitor the Chebyshev tail

$$\eta_{\rm tail} = \frac{\max_{N-15\le k\le N}|c_k|}{\max_{0\le k\le N}|c_k|}, \quad (57)$$

where c_k are the Chebyshev coefficients of the phase-peeled unknown on each subdomain. Finally, Eqs. (25) and (53) test the physical flux balance after the homogeneous matching and Green integral have both been performed.

The production data below retain the self channel $\ell' = \ell$ when quoting the first-order scattering coefficients. This is not an angular truncation of the first-order field: the cubic Kerr source in Eq. (29) does generate off-diagonal target channels. However, at order ϵ the total energy flux contains only interference between $\Phi^{(0)}$ and $\Phi^{(1)}$. Orthogonality of the spheroidal harmonics then removes every target channel except $\ell' = \ell$ from the flux correction. Off-diagonal channels are therefore required for the full first-order waveform, but not for the self-channel coefficients $T^{(1)}_{\ell}$ and $R^{(1)}_{\ell}$ reported here.

The angular discretization is inexpensive enough to audit independently of the radial Green calculation. Table I tests both the largest spheroidal parameter in the displayed axisymmetric data, $(\ell, m, M\omega; \ell') = (2, 0, 1.0; 2)$, and, as a separate angular-only stress test, the projector $(2, 2, 0.30; 6)$. The latter does not use a nonaxisymmetric radial solution and is not a nonlinear target-channel flux measurement. The active angular settings change the separation constants by at most 4.4×10^{-16} and the source projectors by at most 5.3×10^{-14} relative to the reference. Lowering the basis and quadrature separately gives changes of the same order or less, so angular discretization is not the limiting error in the reported data.

Local field diagnostic. At the representative peak in Fig. 1, the 600-point Chebyshev–Gauss grid on each of the three homogeneous branches gives 1800 independent samples. Across the 1800 off-grid validation points, the field-dependent radial residual has maximum, 95th-percentile, and median values 2.8×10^{-10}, 1.2×10^{-11},

TABLE I. Angular-discretization audit. Each entry is the largest relative change over $(\ell, m, M\omega; \ell') = (2, 0, 1.0; 2)$ and $(2, 2, 0.30; 6)$, measured against the reference $(L_{\rm extra}, q_\theta) = (24, 512)$.

setting $(L_{\rm extra}, q_\theta)$	$\max \Delta\lambda$	$\max \Delta C^{(0)}$	$\max \Delta C^{(2)}$
active $(16, 256)$	4.4×10^{-16}	3.7×10^{-14}	5.2×10^{-14}
basis $(12, 512)$	4.4×10^{-16}	1.6×10^{-16}	0
quadrature $(24, 128)$	0	1.5×10^{-14}	2.0×10^{-14}

and 3.2×10^{-13}. The largest normalized Chebyshev tail is 2.8×10^{-14} and the Wronskian mismatch is below 3×10^{-16}. These local diagnostics are reported separately from the global flux-balance residual.

C. Resonance and high-frequency diagnostics

The nonlinear boundary amplitude combines the scattering Green function and the frequency-dependent source. Write the cubic self-channel coefficient as $A_H^{(1)} = -\mathcal{I}_H/W$, with $\mathcal{I}_H = \int_{r_+}^{\infty} R^{\rm up} Q_{\ell\ell m}\, dr$. The QNM frequencies are zeros of the analytically continued homogeneous Wronskian. On the real axis, the flux response also contains the source projection, its phase, and the fixed-incident normalization in Eq. (46). This structure motivates a Breit–Wigner-type description of a resolved peak and allows its center to shift relative to the QNM real part. We measure the peak position and width from the flux curve and assess their stability under changes of the fitting window.

For the high-frequency analysis, the real scalar radial operator at real $\omega > 0$ gives $R^{\rm down} = \overline{R^{\rm up}}$ and

$$R^{\rm in} = B^{\rm inc}\overline{R^{\rm up}} + B^{\rm ref} R^{\rm up},$$
$$W = 2i\omega B^{\rm inc}. \tag{58}$$

Let $P(r) = r^2 C^{(0)}_{\ell\ell m} + a^2 C^{(2)}_{\ell\ell m}$, which is real, and define the source projections

$$J_{\rm d} = \int_{r_+}^{\infty} P |R^{\rm in}|^2 |R^{\rm up}|^2\, dr,$$
$$J_{\rm o} = \int_{r_+}^{\infty} P |R^{\rm in}|^2 (R^{\rm up})^2\, dr. \tag{59}$$

Substitution into the Green integral gives the exact decomposition

$$A_H^{(1)} = -\frac{1}{2i\omega}\left[J_{\rm d} + \frac{B^{\rm ref}}{B^{\rm inc}} J_{\rm o}\right],$$
$$\chi_H = -\frac{{\rm Im}[(B^{\rm ref}/B^{\rm inc}) J_{\rm o}]}{\omega |B^{\rm inc}|^2}. \tag{60}$$

The real integral $J_{\rm d}$ contributes a horizon-phase shift. The absorbed-flux correction is the interference projection involving the linear reflection amplitude and the oscillatory integral $J_{\rm o}$. Thus its tail depends on their combined magnitude and relative phase. This is the Kerr counterpart of the decomposition in Sec. III.B.3 of the Schwarzschild reference[18], with the Kerr radial functions and covariant weight retained explicitly.

For the cubic Schwarzschild calculation, that reference obtains a high-frequency envelope proportional to $(M\omega)^{-1/2} \exp(-\omega/T_H)$ in its stated normalization and source-sign convention. This motivates testing

$$T_\ell^{(1)} \simeq A_\ell (M\omega)^{-\alpha} \exp(-\omega/T_{\rm fit}), \qquad \alpha = 0, \tfrac{1}{2}, \tag{61}$$

on the same resolved Kerr samples. The Kerr decay exponent requires control of both factors in Eq. (60) and their complex-radial continuation. The available real-frequency calculations test the finite-window models and their predictive residuals.

At high frequencies the phase-peeled homogeneous amplitudes become slowly varying, while the nonlinear Green integral is controlled by rapidly oscillatory tails. Motivated by the thermal scale associated with a black-hole horizon [25] and by finite-window Boltzmann diagnostics in the Kerr graybody-factor literature [26, 27], we test an exponential finite-window diagnostic using the Kerr radial normalization and the Σ-weighted source. The baseline model is

$$T_\ell^{(1)}(\omega) \simeq A_\ell \exp\left[-\frac{\omega}{T_{\rm fit}}\right], \qquad \omega > {\rm Re}\,\omega_{\ell 0}. \tag{62}$$

For Kerr

$$T_H = \frac{r_+ - r_-}{4\pi(r_+^2 + a^2)}. \tag{63}$$

For $a = 0.5$ the fits in Sec. IV give $\pi M T_{\rm fit} = 0.111$, 0.105, and 0.101 for $\ell = 0, 1, 2$, close to $\pi M T_H = 0.1160$. These values should be read as finite-window effective temperatures. Their dependence on an algebraic prefactor is tested in Sec. IV A 2.

IV. RESULTS

A. Resonant cubic response and the post-resonance tail

The finite-frequency cubic response is shown in Fig. 2. It plots $T_\ell^{(1)}$ for $\ell = 0, 1, 2$, uses a linear scale to display the resonance peaks, and uses a logarithmic scale to display the high-frequency tail. The dashed lines mark the real parts of the fundamental scalar Kerr quasinormal modes computed with the local `qnm` package [28] for $s = 0$, $a = 0.5$, $m = 0$, and $n = 0$.

Three features are robust in the accepted data. First, in the axisymmetric Kerr cases studied here, $T_\ell^{(1)}$ is positive at every accepted sampled frequency in the present flux convention. For positive coupling in Eq. (5), the

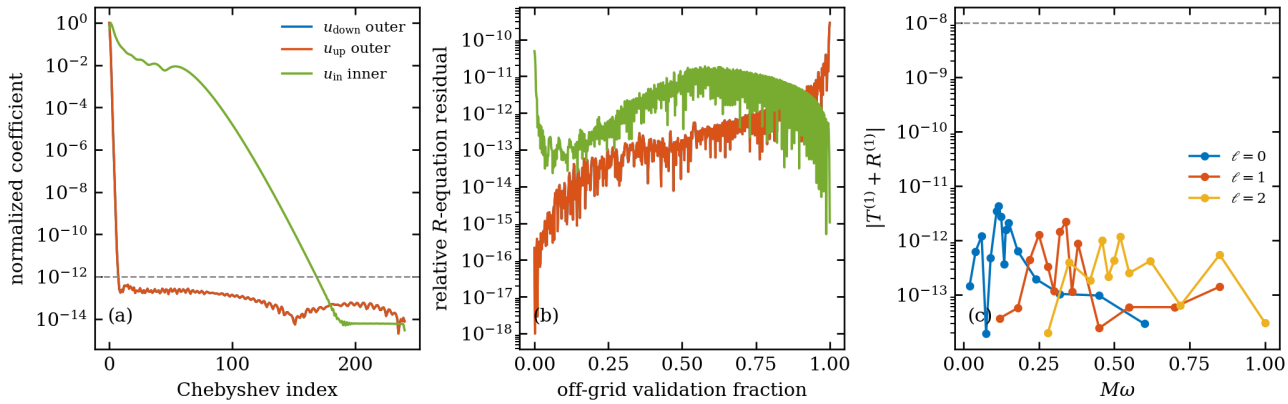


FIG. 1. Internal numerical accuracy diagnostics for the axisymmetric Kerr calculation. Panels (a) and (b) use the representative peak point $a = 0.5$, $m = 0$, $\ell = 2$, and $M\omega = 0.52$ with $N = 240$ and quadrature order $q = 192$. Panel (a) shows the normalized Chebyshev coefficients of the phase-peeled homogeneous solutions on the outer and inner subdomains. Panel (b) shows the relative residual (56) after evaluating the spectral interpolant and its analytic derivatives on 600 off-grid Chebyshev–Gauss points per branch in the original radial Teukolsky equation. Panel (c) shows the first-order flux-balance residual for the retained axisymmetric points. The dashed line marks the acceptance threshold 10^{-8} used before any point enters the manuscript figures or fits.

nonlinear correction therefore increases the absorbed flux and, by Eq. (53), decreases the reflected flux at the same order. This sign statement refers to the coefficient convention of Eqs. (46) and (47); it should not be compared to a reduced Schwarzschild field convention without translating the flux normalization.

1. *Peak locations and source-weighted resonance*

The peaks of $T_\ell^{(1)}$ lie on the frequency scale set by the real parts of the corresponding fundamental Kerr scalar QNMs [4, 28]. The fitted peak positions are $M\omega_{\rm peak} = 0.1306$, 0.3101, and 0.5026 for $\ell = 0, 1, 2$, while the QNM real parts are 0.1124, 0.2979, and 0.4920. The corresponding fractional offsets are 16.2%, 4.1%, and 2.2%. We characterize the real-frequency resonance by fitting the selected samples to the four-parameter profile

$$T_\ell^{(1)}(\omega) = T_{\rm off} + \frac{A\,\gamma^2}{(M\omega - \omega_p)^2 + \gamma^2}. \tag{64}$$

Here ω_p is the fitted center in units $M = 1$, and γ is the half-width parameter of the phenomenological Breit–Wigner-type interpolation. The coefficient of determination of the three fits is $R^2 = 0.9918$, 0.9978, and 0.9985 for $\ell = 0, 1, 2$, respectively. Equation (60) and the full Green integral identify the flux as a source-weighted interference response. Its fitted peak samples the QNM frequency scale together with the frequency dependence of the nonlinear source and incident normalization. A complex-frequency QNM projection requires a conserved bilinear form and contour regularization [29]. The fit uses the sampled points satisfying $T_\ell^{(1)} > 0.2T_{\ell,\max}^{(1)}$; the relative RMS residuals of these peak fits over the selected points are 2.41%, 1.27%, and 1.05% for $\ell = 0, 1, 2$, respectively. These residuals quantify the descriptive quality of the real-frequency interpolation. As a fit-window sensitivity check, we repeat the selection with thresholds from 0.15 to 0.30 of the sampled maximum, retaining at least five points in every fit. The resulting fitted centers vary by at most 6.7×10^{-4}, 3.9×10^{-4}, and 3.4×10^{-4} for $\ell = 0, 1, 2$, respectively, whereas the corresponding half-widths vary by 3.1×10^{-3}, 4.8×10^{-3}, and 5.2×10^{-3}. The center locations are therefore stable at the resolution of the sweep, while the widths should be regarded as descriptive fit parameters. The complete threshold scan is included in the reproducibility data.

2. *High-frequency decay and its effective scale*

At high frequency the accepted tail points are well fit by the Boltzmann form Eq. (62). The fitted temperatures differ from the Kerr Hawking temperature by 4.0%, 9.8%, and 12.8% for $\ell = 0, 1, 2$, respectively. These are effective finite-window fit parameters within the stated post-resonance windows. On the accepted low-frequency side the Kerr fixed-incident coefficients decrease as ω is lowered over the globally balanced windows. The deeper horizon-side data then recover the matched Kerr powers $(M\omega)^{2\ell+2}$ with free-fit $R^2 > 0.999989$. The comparison uses the fixed-incident normalization; the translation to the reduced Schwarzschild convention is given in Appendix A.

The exponential fits use unweighted least squares for $\log T_\ell^{(1)}$ versus $M\omega$ over the windows $[0.18, 0.60]$, $[0.38, 0.85]$, and $[0.62, 1.00]$ for $\ell = 0, 1, 2$, respectively. The corresponding root-mean-square residuals in log space are 3.7×10^{-2}, 6.7×10^{-2}, and 5.6×10^{-2}. These

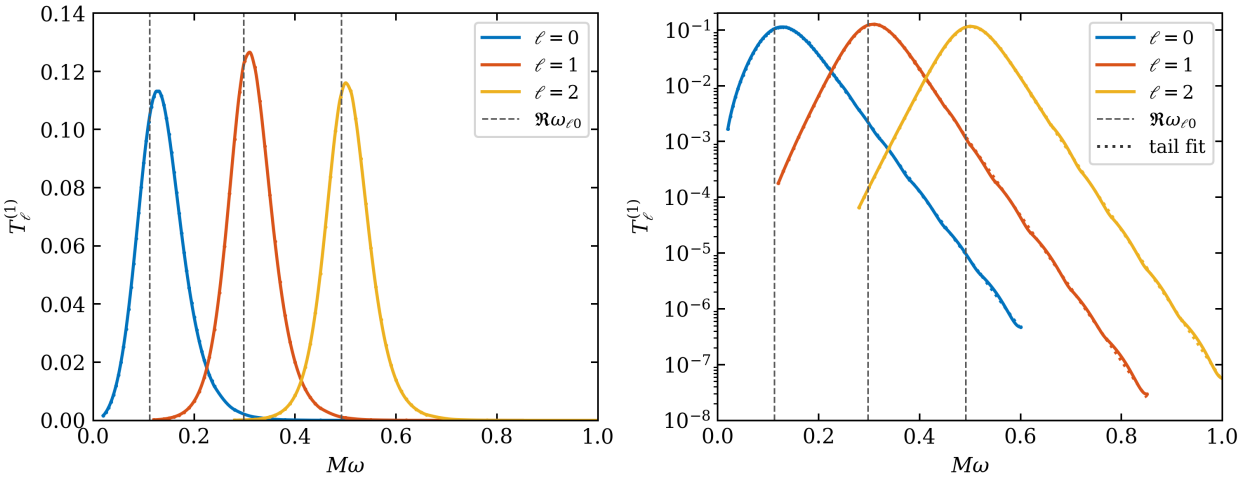


FIG. 2. Nonlinear Kerr transmission coefficients $T_\ell^{(1)}$ for $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$. The left panel uses a linear scale to show the Breit–Wigner-type peaks. The right panel uses a logarithmic scale to show the exponential high-frequency tails; dotted segments show the finite-window fits used in Table II. Dashed vertical lines mark the real parts of the fundamental scalar Kerr QNMs $\mathrm{Re}\,\omega_{\ell 0}$. All 43 points entering the three tail-fit windows use $q = 2048$; the denser low-frequency and peak samples use $q = 192$. Circles show every accepted sample, while the solid curves are shape-preserving interpolants used only as visual guides. Every plotted point satisfies $|T^{(1)} + R^{(1)}| \leq 10^{-8}$.

TABLE II. Fitting parameters of the Kerr nonlinear transmission resonance and the high-frequency Boltzmann tail. The QNM frequencies are scalar Kerr fundamental modes with $s = 0$, $a = 0.5$, $m = 0$, and $n = 0$. All frequencies are reported in units $M = 1$.

ℓ	ω_p	γ	$\Re\omega_Q$	$\lvert\Im\omega_Q\rvert$	πT_{fit}	ΔT
0	0.1306	0.0599	0.1124	0.1022	0.1114	4.00%
1	0.3101	0.0591	0.2979	0.0954	0.1046	9.85%
2	0.5026	0.0595	0.4920	0.0946	0.1012	12.81%

window-dependent residuals quantify the quality of the finite-window fit; they are not estimates of a systematic error outside the computed frequency range.

Because the effective temperature is a fit diagnostic rather than an analytic observable, we also repeat the tail fit after shifting either edge of each window while retaining at least three positive data points. The resulting window ranges and the largest logarithmic fit residual are given in Table III. The fitted values vary only at the percent level within these local windows, while their absolute offsets from T_H remain consistent with the finite-window qualification above.

To test whether a power law can mimic the exponential over these finite windows, we fit the same positive data to $\log T^{(1)} = c - \beta M\omega$ and $\log T^{(1)} = c - n\log(M\omega)$. Both models have two fitted parameters, so their corrected Akaike information criteria can be compared without a parameter-count ambiguity. For n_s samples and $k = 2$ fitted parameters we use

$$\mathrm{AICc} = n_s \log(\mathrm{RSS}/n_s) + 2k + \frac{2k(k+1)}{n_s - k - 1}, \tag{65}$$

with the residual sum of squares evaluated in log space. Table V also reports leave-one-out prediction residuals. The exponential has the smaller AICc and the smaller predictive residual for every ℓ. A fit with the exponential slope fixed exactly to $-1/T_H$ has logarithmic RMS residuals 0.139, 0.443, and 0.514 for $\ell = 0, 1, 2$; hence the data support a finite-window exponential more strongly than either a power law or an exact thermal slope. This comparison establishes exponential decay as the preferred description within the resolved windows. Extending the frequency interval tests its asymptotic persistence.

3. *Algebraic prefactor and the Schwarzschild comparison*

We next fit Eq. (61) with $\alpha = 1/2$ to the same 19, 14, 10 samples used by the baseline exponential. Both models have two fitted parameters, the amplitude and the decay scale. Table IV reports their logarithmic residuals, leave-one-out prediction errors, and AICc differences. The prefactor improves the residuals for $\ell = 1, 2$, with the stronger preference at $\ell = 1$; for $\ell = 0$ the baseline exponential has the smaller AICc and the two predictive errors are nearly equal. These channel-dependent comparisons quantify how an algebraic factor changes the decay scale in the measured finite-frequency regime.

The logarithmic slope of the prefactor model satisfies

$$-\frac{d\log T^{(1)}}{d\omega} = \frac{1}{T_{\mathrm{fit}}} + \frac{\alpha}{\omega}. \tag{66}$$

A pure exponential fit therefore incorporates the algebraic contribution into its effective inverse temperature. Including $\alpha = 1/2$ raises $\pi M T_{\mathrm{fit}}$ from

TABLE III. Sensitivity of the finite-window effective temperatures to the fit window. The baseline window is the one used in Table II; the range is obtained from three nearby edge-shifted windows. The final column is the largest absolute fractional offset from the Kerr Hawking temperature among the four windows.

ℓ	baseline $[M\omega_{\min}, M\omega_{\max}]$	$\pi T_{\rm fit}^{\rm base}$	range of $\pi T_{\rm fit}$	$\max \sigma_{\log}$	$\max \lvert T_{\rm fit}/T_H - 1\rvert$
0	$[0.18, 0.60]$	0.1114	$[0.1109, 0.1122]$	3.86×10^{-2}	4.45%
1	$[0.38, 0.85]$	0.1046	$[0.1037, 0.1068]$	6.95×10^{-2}	10.61%
2	$[0.62, 1.00]$	0.1012	$[0.1002, 0.1031]$	5.59×10^{-2}	13.62%

TABLE IV. Schwarzschild-inspired prefactor test on the same Kerr tail windows as Table V. The fitted model is $A(M\omega)^{-1/2}\exp(-\omega/T_{1/2})$. ΔAICc = AICc$_{1/2}$ − AICc$_{\rm exp}$; negative values favor the prefactor model. The last column fixes its temperature to the Kerr T_H and fits the amplitude alone. All residuals are RMS errors in $\log T^{(1)}$.

ℓ	$\pi M T_{1/2}$	$\sigma_{1/2}$	$\sigma_{\rm LOO,1/2}$	ΔAICc	$\sigma_{1/2,T_H}$
0	0.11754	0.04186	0.04572	+4.48	0.05893
1	0.10772	0.05505	0.07332	−5.53	0.31412
2	0.10328	0.05094	0.07388	−1.85	0.43161

$0.11139, 0.10460, 0.10116$ to $0.11754, 0.10772, 0.10328$ for $\ell = 0, 1, 2$. The revised scale lies within 1.3% of T_H for the monopole. The exact-T_H prefactor model has RMS residuals $0.05893, 0.31412, 0.43161$; the free-scale model resolves the remaining modal dependence. The source weight, reflection amplitude, and phase in Eq. (60) supply the quantities whose high-frequency asymptotics determine that dependence. This connects the numerical tail to the Schwarzschild analytic argument while retaining the Kerr-specific ingredients.

All peak and tail fits, and all filled sweep points in Figs. 4 and 2, use only the accepted globally balanced subset. The open infrared points in Fig. 4 instead obey the paired horizon-side and Wronskian checks stated above. The largest first-order balance residual of the global subset is 6.9×10^{-12}, 2.2×10^{-12}, and 1.5×10^{-12} for $\ell = 0, 1, 2$, respectively. Completed solver rows that fail the 10^{-8} gate remain in the reproducibility data as an audit trail, but they are omitted from the plots, tables, peak interpolation, and low-frequency discussion. No physical claim is based on those excluded rows.

The change from the former production resolution to the active refined resolution is quantified separately in Table VII. At the representative peak points the refined calculation changes the transmission coefficient by at most 3.2×10^{-5} and preserves the first-order flux balance at the 10^{-12} level. We therefore use the refined curves for the physical discussion below; the lower-resolution values are reported to make the resolution choice auditable rather than folded into an unreported systematic error.

The high-frequency tail requires a separate quadrature audit because flux balance alone does not detect a common error in the two interference terms. At fixed $N = 240$, the compact inner integral was scanned from $q = 192$ to $q = 2048$, with twelve intermediate orders, at the three representative points in Table VIII. The sequence is mildly nonmonotonic at intermediate order but reaches a plateau at the two largest orders. We therefore recomputed every point entering the tail fits at $q = 2048$. Lowering N at fixed $q = 2048$ produces a much smaller change, showing that the remaining sensitivity belongs to the oscillatory Green integral rather than to the homogeneous spectral solution.

B. Infrared response and local-power hierarchy

The scalar monopole provides an analytic low-frequency normalization check directly in the Kerr Teukolsky variable. Fix $|a| < M$, take $(\ell, m) = (0, 0)$, and introduce the far-zone coordinate $x = \omega r$. In the limit $\omega \to 0$ the normalized spheroidal harmonic and the self-channel projector become

$$S_{00} \longrightarrow \frac{1}{\sqrt{4\pi}}, \qquad C^{(0)}_{000} \longrightarrow \frac{1}{4\pi}. \tag{67}$$

The leading far-zone radial equation is the flat-space monopole equation. Matching to the solution that is unity at the horizon gives

$$\begin{aligned} R^{\rm in}_{00\omega} &= j_0(x) + o(1) = \frac{\sin x}{x} + o(1), \\ R^{\rm up}_{00\omega} &= \frac{e^{ix}}{r}[1 + o(1)]. \end{aligned} \tag{68}$$

Comparison with Eq. (22) and evaluation of Eq. (41) at infinity then yield

$$B^{\rm inc}_{00\omega} = \frac{\mathrm{i}}{2\omega}[1 + o(1)], \qquad W_0 = 2\mathrm{i}\omega B^{\rm inc}_{00\omega} = -1 + o(1). \tag{69}$$

The $a^2 C^{(2)}$ part of the source and all strong-field corrections are subleading relative to the far-zone $r^2 C^{(0)}$ term. Substitution of Eqs. (67)–(69) into the horizon

TABLE V. Finite-window comparison of exponential and power-law descriptions of the high-frequency nonlinear transmission tail. The residuals are RMS errors in $\log T^{(1)}$; ΔAICc is $\mathrm{AICc}_{\mathrm{power}} - \mathrm{AICc}_{\mathrm{exp}}$, so positive values favor the exponential. The final two columns are leave-one-out logarithmic prediction residuals.

ℓ	points	σ_{exp}	σ_{power}	ΔAICc	$\sigma_{\mathrm{LOO,exp}}$	$\sigma_{\mathrm{LOO,power}}$
0	19	3.72×10^{-2}	4.89×10^{-1}	97.9	4.56×10^{-2}	6.09×10^{-1}
1	14	6.71×10^{-2}	3.75×10^{-1}	48.2	8.92×10^{-2}	4.80×10^{-1}
2	10	5.59×10^{-2}	1.84×10^{-1}	23.8	8.11×10^{-2}	2.62×10^{-1}

TABLE VI. Numerical quality of the axisymmetric Kerr frequency sweeps. Here $\delta_0 = |T^{(0)} + R^{(0)} - 1|$ and $\delta_1 = |T^{(1)} + R^{(1)}|$. Every retained point satisfies $\delta_1 \le 10^{-8}$. All active points use $N = 240$. The low-frequency and peak samples use $q = 192$. All 43 samples in the three tail-fit windows use $q = 2048$ after the audit in Table VIII. The earlier $N = 224$, $q = 160$ peak sweep is retained as the production-resolution audit in Table VII. The final column is the root-mean-square residual of the logarithmic high-frequency tail fit.

ℓ	retained	$M\omega_{\mathrm{min}}$	$\max \delta_0$	$\max \delta_1$	σ_{tail}
0	51	0.02	2.8×10^{-11}	6.9×10^{-12}	3.7×10^{-2}
1	40	0.12	1.9×10^{-11}	2.2×10^{-12}	6.7×10^{-2}
2	44	0.28	3.8×10^{-11}	1.5×10^{-12}	5.6×10^{-2}

Green integral therefore gives the full complex amplitude

$$\begin{aligned} A^{(1)}_{H,00} &= \frac{C^{(0)}_{000}}{\omega^2} \int_0^\infty \frac{\sin^3 x\, e^{\mathrm{i}x}}{x^2}\, \mathrm{d}x + o(\omega^{-2}) \\ &= \frac{C^{(0)}_{000}}{\omega^2} (I_0 + \mathrm{i} J_0) + o(\omega^{-2}), \end{aligned} \tag{70}$$

where both integrals can be evaluated without a regulator. Using $\sin^3 x \cos x = \frac{1}{4}\sin 2x - \frac{1}{8}\sin 4x$ and integrating by parts,

$$\begin{aligned} I_0 &= \int_0^\infty \frac{\sin^3 x \cos x}{x^2}\, \mathrm{d}x \\ &= \frac{1}{2}\int_0^\infty \frac{\cos 2x - \cos 4x}{x}\, \mathrm{d}x = \frac{1}{2}\ln 2. \end{aligned} \tag{71}$$

while a second integration by parts gives

$$J_0 = \int_0^\infty \frac{\sin^4 x}{x^2}\, \mathrm{d}x = 4\int_0^\infty \frac{\sin^3 x \cos x}{x}\, \mathrm{d}x = \frac{\pi}{4}. \tag{72}$$

Since $C^{(0)}_{000} = 1/(4\pi)$, the amplitude and relative response are

$$\begin{aligned} A^{(1)}_{H,00} &= \left(\frac{\ln 2}{8\pi} + \frac{\mathrm{i}}{16}\right)\omega^{-2} + o(\omega^{-2}), \\ \rho_{H,00} &\longrightarrow \frac{\ln 2}{2\pi} + \frac{\mathrm{i}}{4}. \end{aligned} \tag{73}$$

The imaginary term is a finite horizon phase response after incident normalization; it does not contribute to the first-order energy flux. The real part and Eq. (48) give

$$\boxed{\lim_{\omega\to 0} \chi_{H,00} = 8 C^{(0)}_{000} I_0 = \frac{\ln 2}{\pi} = 0.220635600153.} \tag{74}$$

The leading result is independent of spin because the limit is controlled by the far zone. The closed form $\ln 2/\pi$ is specific to the scalar monopole and is not a superradiant-threshold result: $m = 0$ implies $p = \omega$.

The same matching argument determines both absolute flux coefficients for every cubic axisymmetric self channel. Let $d = (M^2 - a^2)^{1/2}$ and first set $\omega = 0$. The horizon-regular radial equation and its unit-horizon solution are

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta \frac{\mathrm{d}R}{\mathrm{d}r}\right) - \ell(\ell+1)R = 0, \qquad R = P_\ell\!\left(\frac{r-M}{d}\right), \tag{75}$$

because $P_\ell(1) = 1$. Since the leading coefficient of $P_\ell(z)$ is $q_\ell = (2\ell)!/[2^\ell(\ell!)^2]$, matching the growing part of Eq. (75) to the small-x limit of $j_\ell(x)$ gives

$$\begin{aligned} R^{\mathrm{in}}_{\ell 0\omega} &= \mathcal{K}_\ell (M\omega)^{-\ell} j_\ell(x)[1 + o(1)], \\ \mathcal{K}_\ell &= (2\ell+1)!!\, q_\ell \left(\frac{M}{d}\right)^\ell, \qquad x = \omega r. \end{aligned} \tag{76}$$

The incoming part of j_ℓ fixes both the phase and magnitude,

$$\begin{aligned} B^{\mathrm{inc}}_{\ell 0\omega} &= \frac{M \mathrm{i}^{\ell+1} \mathcal{K}_\ell}{2(M\omega)^{\ell+1}}[1 + o(1)], \\ |B^{\mathrm{inc}}_{\ell 0\omega}|^2 &= \frac{M^2 \mathcal{K}_\ell^2}{4(M\omega)^{2\ell+2}}[1 + o(1)]. \end{aligned} \tag{77}$$

Substitution into Eq. (23), with $p = \omega$, yields the absolute linear law

$$\begin{aligned} T^{(0)}_{\ell 0} &= \mathcal{A}^{(0)}_\ell (M\omega)^{2\ell+2}[1 + o(1)], \\ \mathcal{A}^{(0)}_\ell &= \frac{4(r_+^2 + a^2)}{M^2 \mathcal{K}_\ell^2}. \end{aligned} \tag{78}$$

TABLE VII. Production-resolution audit for the axisymmetric Kerr calculation. The entries compare the former production configuration $(N, q) = (224, 160)$ with the active reference configuration $(N, q) = (240, 192)$ at fixed $r_{\rm m} = 40$. Each entry in the first four columns is $|X_{240,192} - X_{224,160}|/|X_{240,192}|$ for the indicated Green-function amplitude or flux coefficient; the final column is the absolute first-order balance residual of the reference run.

case	$\Delta A_{\rm ref}$	ΔA_H	$\Delta T^{(1)}$	$\Delta R^{(1)}$	$\lvert\delta_1\rvert_{\rm ref}$
$\ell = 0,\ M\omega = .125$	4.5×10^{-7}	7.8×10^{-7}	2.3×10^{-6}	2.3×10^{-6}	2.7×10^{-12}
$\ell = 1,\ M\omega = .34$	5.9×10^{-6}	3.3×10^{-6}	1.5×10^{-5}	1.5×10^{-5}	2.2×10^{-12}
$\ell = 2,\ M\omega = .52$	9.6×10^{-6}	7.6×10^{-6}	3.1×10^{-5}	3.1×10^{-5}	1.2×10^{-12}

TABLE VIII. High-frequency convergence audit for the active tail data. The quadrature columns compare $q = 1792$ with the $q = 2048$ reference at $N = 240$; the radial-order column compares $N = 224$ with $N = 240$ at $q = 2048$. All entries except the final balance residual are relative changes with respect to the reference.

case	$\Delta T^{(1)}[q]$	$\Delta T^{(1)}[N]$	$\Delta A_{\rm ref}[q]$	$\Delta A_H[q]$	$\lvert\delta_1\rvert_{\rm ref}$
$\ell = 0,\ M\omega = .52$	3.0×10^{-5}	7.1×10^{-10}	2.3×10^{-5}	1.6×10^{-9}	7.8×10^{-15}
$\ell = 1,\ M\omega = .77$	1.7×10^{-4}	1.0×10^{-9}	3.1×10^{-4}	1.2×10^{-9}	2.7×10^{-13}
$\ell = 2,\ M\omega = .92$	2.5×10^{-4}	9.5×10^{-8}	3.4×10^{-4}	2.5×10^{-9}	1.0×10^{-14}

For the nonlinear coefficient, define the spherical angular overlap and two convergent Bessel integrals

$$\mathcal{C}_\ell = \int |Y_{\ell 0}|^4 \mathrm{d}\Omega, \qquad \mathcal{I}_\ell = -\int_0^\infty x^2 y_\ell(x) j_\ell^3(x) \mathrm{d}x,$$
$$\mathcal{J}_\ell = \int_0^\infty x^2 j_\ell^4(x) \mathrm{d}x > 0. \tag{79}$$

For fixed subextremal spin the $a^2 C^{(2)}$ source term and the strong-field part of the Green integral are subleading. In units $M = 1$, the outgoing solution and Wronskian in the far zone are $R^{\rm up} = \omega \mathrm{i}^{\ell+1} h_\ell^{(1)}(x)[1 + o(1)]$ and $W_\ell = \mathrm{i}^{\ell+2} \mathcal{K}_\ell \omega^{-\ell}[1 + o(1)]$. Their phases are essential. Since $\mathrm{i}h_\ell^{(1)} = \mathrm{i}j_\ell - y_\ell$, direct substitution in the Green integral gives, after restoring M,

$$\frac{A^{(1)}_{H,\ell 0}}{M^2} = \mathcal{C}_\ell \mathcal{K}_\ell^2 (\mathcal{I}_\ell + \mathrm{i}\mathcal{J}_\ell)(M\omega)^{-2\ell-2}[1 + o(1)]. \tag{80}$$

Thus the unit-horizon coefficient has $|A_H^{(1)}| \propto (M\omega)^{-2\ell-2}$ and $\arg A_H^{(1)} \to \arg(\mathcal{I}_\ell + \mathrm{i}\mathcal{J}_\ell)$. This divergence is not a divergent scattering response: maintaining unit horizon amplitude requires a growing incident wave as $\omega \to 0$. Combining Eqs. (77) and (80) instead gives

$$M A^{(1)}_{H,\mathrm{fix}} = \frac{8\mathcal{C}_\ell}{\mathcal{K}_\ell} \mathrm{i}^{-(\ell+1)} (\mathcal{I}_\ell + \mathrm{i}\mathcal{J}_\ell)(M\omega)^{\ell+1}[1 + o(1)],$$
$$\rho_{H,\ell 0} \longrightarrow \rho^{\rm IR}_{H,\ell} = 4\mathcal{C}_\ell (\mathcal{I}_\ell + \mathrm{i}\mathcal{J}_\ell). \tag{81}$$

The fixed-incident first-order amplitude therefore vanishes as $(M\omega)^{\ell+1}$, while its ratio to the linear horizon amplitude tends to a finite complex constant. Taking its real part cancels the near-zone matching coefficient and recovers the flux laws

$$\chi_{H,\ell 0} \longrightarrow \chi_\ell^{\rm IR} = 2\,\mathrm{Re}\,\rho^{\rm IR}_{H,\ell} = 8\mathcal{C}_\ell \mathcal{I}_\ell,$$
$$T^{(1)}_{\ell 0} = \mathcal{A}^{(1)}_\ell (M\omega)^{2\ell+2}[1 + o(1)], \qquad \mathcal{A}^{(1)}_\ell = \mathcal{A}^{(0)}_\ell \chi_\ell^{\rm IR}. \tag{82}$$

Thus the cubic fractional response is unsuppressed in the infrared for every axisymmetric self channel: it changes the coefficient, but not the linear absorption power. The ratio $\chi_\ell^{\rm IR}$ is independent of spin, although the absolute coefficients retain spin dependence through $\mathcal{K}_\ell$ and the horizon-area factor. This statement takes $\omega \to 0$ at fixed $|a| < M$; Eq. (75) also makes clear why the limit is not uniform as $a/M \to 1$ for $\ell > 0$.

Table IX evaluates the new formulas for the $a/M = 0.5$ production spin. The Bessel integrals were summed over successive π intervals and extrapolated in inverse endpoint powers. Independent Kerr solves at $(N, q) = (144, 144)$ and $(176, 176)$ used only the horizon-side response at the listed lowest frequencies. At the higher resolution the relative differences between the scaled numerical and analytic coefficients are $(0.48, 0.96, 3.32)\%$ for $T^{(0)}$ and $(0.82, 1.07, 3.41)\%$ for $T^{(1)}$, ordered by $\ell = 0, 1, 2$; these are finite-frequency corrections, not fitted exponents. The two-resolution changes in χ_H are given in the final column, and the largest Wronskian error is 2.5×10^{-11}. The same runs test information absent from the energy flux. For $\ell = 0, 1, 2$ they give $\mathcal{J}_\ell = (0.785398163397, 0.366519142919, 0.246839422782)$ and limiting phases $(66.1895°, 70.6933°, 72.4699°)$. At the lowest listed frequencies the complex relative responses agree with Eq. (81) to $(0.228, 0.148, 0.318)\%$, while their simultaneous-resolution changes are $(6.3 \times 10^{-6}, 5.0 \times 10^{-5}, 3.0 \times 10^{-4})$. Hence both the dissipative real part and the reactive phase part are independently resolved. The infinity-side $R^{(1)}$ interference term is already cancellation limited at these frequencies and is deliberately not used in this infrared audit.

As a numerical test of the powers rather than only their coefficients, we fit the seven horizon-side points for each multipole freely in log–log space. For $T^{(0)}$ the fitted powers are 2.0145, 4.0298, and 6.0784; for $T^{(1)}$ they are 2.0245, 4.0360, and 6.0884. Every fit

has $R^2 > 0.999989$. Restricting to the lowest five frequencies moves the two sets to $(2.0095, 4.0193, 6.0502)$ and $(2.0162, 4.0237, 6.0585)$, respectively, showing convergence toward the exact powers $(2, 4, 6)$ as the upper end of the fit window is removed. The analytic conclusion still comes from matched asymptotics; these free fits are an independent finite-frequency audit. For $\ell = 2$, attempts at $M\omega = 0.003$ and 0.004 failed the paired continuation-normalization check and are excluded, so the reported infrared window begins at $M\omega = 0.005$.

This calculation extends analytically to every positive integer n. Consider the $U(1)$-invariant local equation

$$\Box\Phi + \epsilon_n|\Phi|^{2n}\Phi = 0, \qquad n = 1, 2, \ldots, \tag{83}$$

for which $[\epsilon_n] = M^{2n-2}$ when $[\Phi] = M^{-1}$. The cubic model used elsewhere in this paper is the $n = 1$ member. At first order the source scales as $|\alpha|^{2n}\alpha$ under $R^{(0)} \mapsto \alpha R^{(0)}$, so fixed-incident normalization gives

$$\begin{aligned} T_n^{(1)} &= \frac{2p(r_+^2 + a^2)}{\omega|B^{\rm inc}|^{2n+2}} \operatorname{Re} A_{H,n}^{(1)}, \\ \chi_H^{(n)} &\equiv \frac{T_n^{(1)}}{T^{(0)}} = \frac{2\operatorname{Re} A_{H,n}^{(1)}}{|B^{\rm inc}|^{2n}}. \end{aligned} \tag{84}$$

For a general incident mode and target multipole the projected source is

$$Q_{\ell'\ell m}^{(n)}(r) = \left[r^2 C_{\ell'\ell m}^{(0,n)} + a^2 C_{\ell'\ell m}^{(2,n)}\right] |R_{\ell m\omega}^{\rm in}|^{2n} R_{\ell m\omega}^{\rm in}, \tag{85}$$

$$C_{\ell'\ell m}^{(j,n)} = \int \mathrm{d}\Omega\, \cos^j\theta\, S_{\ell' m}^* |S_{\ell m}|^{2n} S_{\ell m}, \qquad j = 0, 2. \tag{86}$$

The Green amplitudes follow from the same in/up Wronskian as in the cubic calculation, with $Q \mapsto Q^{(n)}$. In particular,

$$A_{H,n}^{(1)} = -\frac{1}{W_\ell}\int_{r_+}^{\infty} Q_{\ell\ell m}^{(n)}(r) R_{\ell m\omega}^{\rm up}(r)\mathrm{d}r. \tag{87}$$

For a dimensionless incident amplitude $\mathcal{A}_{\rm in}$, the self-channel flux expansion and its conservation law are

$$\begin{aligned} T &= T^{(0)} + \epsilon_n|\mathcal{A}_{\rm in}|^{2n} T_n^{(1)} + \mathcal{O}(\epsilon_n^2|\mathcal{A}_{\rm in}|^{4n}), \\ R &= R^{(0)} + \epsilon_n|\mathcal{A}_{\rm in}|^{2n} R_n^{(1)} + \mathcal{O}(\epsilon_n^2|\mathcal{A}_{\rm in}|^{4n}), \\ T_n^{(1)} &+ R_n^{(1)} = 0. \end{aligned} \tag{88}$$

This makes $M^{2n-2}\chi_H^{(n)}$ dimensionless. Numerically, the outer integrand is not formed by raising a cancellation-prone oscillatory sum to a large power. We expand $(B^{\rm inc}e^{-\mathrm{i}\omega r_*} + B^{\rm ref}e^{\mathrm{i}\omega r_*})^{n+1}$ and its conjugate factor binomially, multiply by the target homogeneous wave, and integrate the resulting finite phase channels separately. This gives $2n + 3$ channels with phase indices $q = -2n - 2, -2n, \ldots, 2n + 2$ and asymptotic envelope r^{-2n}. The construction is therefore finite for every fixed positive integer n and retains oscillatory-tail control as the operator order is increased.

For the monopole the angular projections reduce to

$$C_n^{(0)} = \int |S_{00}|^{2n+2}\mathrm{d}\Omega = (4\pi)^{-n}, \qquad C_n^{(2)} = \frac{1}{3}C_n^{(0)}. \tag{89}$$

The leading far-zone Green integral is consequently

$$\begin{aligned} \operatorname{Re} A_{H,n}^{(1)} &= \frac{C_n^{(0)}}{\omega^2} I_n + o(\omega^{-2}), \\ I_n &= \int_0^\infty \frac{\sin^{2n+1} x\cos x}{x^{2n}}\mathrm{d}x, \\ &= \frac{n}{n+1}\int_0^\infty \frac{\sin^{2n+2} x}{x^{2n+1}}\mathrm{d}x > 0. \end{aligned} \tag{90}$$

The second form follows by integration by parts and proves positivity without numerics. Repeated integration by parts, followed by the standard finite cosine expansion of $\sin^{2n+2} x$, gives the regulator-independent finite sum

$$I_n = \frac{(-1)^n}{4(n+1)(2n-1)!}\sum_{j=1}^{n+1}(-1)^{j+1} j^{2n}\binom{2n+2}{n+1-j}\ln j. \tag{91}$$

Using $B^{\rm inc} = \mathrm{i}/(2\omega) + o(\omega^{-1})$ in Eq. (84) yields the all-n infrared law

$$\begin{aligned} M^{2n-2}\chi_{H,00}^{(n)} &= c_n(M\omega)^{2n-2} \\ &\quad + o\big((M\omega)^{2n-2}\big), \\ c_n &= \frac{2I_n}{\pi^n} > 0. \end{aligned} \tag{92}$$

Thus $n = 1$ retains a finite infrared response and reproduces $c_1 = \ln 2/\pi$, whereas every $n > 1$ interaction produces a power-suppressed fractional response in this scattering observable. The next two coefficients are

$$c_2 = 0.0631722552062, \qquad c_3 = 0.0172436384309. \tag{93}$$

For the full $\ell = 0, 1, 2$ spectra, the far-zone normalization and the covariant source weight $r^2 C_{\ell\ell m}^{(0)} + a^2 C_{\ell\ell m}^{(2)}$ still control how quickly a finite-frequency calculation reaches Eqs. (78) and (82). We therefore retain a separate globally balanced Kerr experiment for $a = 0.5$ and $m = 0$, rather than use the asymptotic formulas to manufacture display points.

The infinity-side interference integral becomes cancellation limited before the transmitted coefficient loses smoothness. For points that test both boundary fluxes, we therefore apply the same global requirement used throughout the paper,

$$|\delta_1| = |T^{(1)} + R^{(1)}| \le 10^{-8}, \tag{94}$$

before a point can enter the globally balanced part of a figure or a two-boundary physical fit. Table X gives the resulting displayed windows. The lowest retained frequencies are $M\omega = 0.02$, 0.12, and 0.28 for $\ell = 0, 1, 2$, respectively. The filled data in Fig. 4 show that both $T^{(0)}$ and $T^{(1)}$ decrease rapidly toward the low-frequency side.

TABLE IX. Analytic cubic infrared coefficients and direct horizon-side checks for $a/M = 0.5$. Here $T^{(i)}_{\ell 0} \sim \mathcal{A}^{(i)}_\ell (M\omega)^{2\ell+2}$ and $\delta_N \chi_H$ compares the simultaneous $(N, q) = (144, 144)$ and $(176, 176)$ calculations at the lowest listed frequency.

ℓ	$\mathcal{I}_\ell$	$\chi^{\rm IR}_\ell$	$\mathcal{A}^{(0)}_\ell$	$\mathcal{A}^{(1)}_\ell$	lowest $M\omega$	$\delta_N \chi_H$
0	0.346573590280	0.220635600153	14.9282032	3.29369308	0.00075	1.1×10^{-7}
1	0.128401008797	0.147136717786	1.24401694	0.183040569	0.00150	1.5×10^{-4}
2	0.077970747421	0.106366541731	0.0165868925	0.00176429039	0.00500	7.7×10^{-5}

The additional open points are the paired horizon-side calculations of Table IX; they extend below the reflected-channel cancellation boundary and are compared with the analytic leading laws. We do not fit a power law to the globally balanced segments. The free infrared fits quoted above use only the open horizon-side points, while the powers in Eqs. (78) and (82) come from matched asymptotics.

We test each lowest retained point independently. Starting from a $N = 256$, $q = 256$, $r_{\rm m} = 40M$ reference, we lower N to 224, lower q to 192, and move the matching radius to $32M$ and $48M$. Table XI shows that the largest relative change in $T^{(1)}$ is 1.8×10^{-8}; the largest change in $R^{(1)}$ is 2.0×10^{-8}, and every run satisfies the first-order balance at or below 3.1×10^{-12}. The displayed low-frequency-side trend is therefore supported by both boundary fluxes, rather than inferred from a cancellation-limited reflected amplitude.

We next test the all-n prediction in Eq. (92) without using it as an input to the radial solve. For $n = 2, 3$ we computed the $\ell = m = 0$ response at $M\omega = (0.0005, 0.00075, 0.001, 0.0015, 0.002, 0.003, 0.005)$ and $a/M = (0, 0.5, 0.9, 0.99, 0.999)$, independently at $(N, q) = (144, 144)$ and $(176, 176)$. A quadratic extrapolation of $\chi^{(n)}_H/\omega^{2n-2}$ over the lowest five frequencies recovers the analytic coefficient at every spin. Changing the fit to the lowest four or six points changes the extrapolated coefficient by at most 4.6×10^{-6}, and the spin-to-spin relative standard deviations of the intercepts are 1.2×10^{-6} and 7.2×10^{-7} for $n = 2, 3$. The direct log–log slopes in the finite fit window are slightly above their asymptotic integer values, with the ranges given in Table XII; their R^2 values exceed 0.9999989.

The fixed-frequency limit does not by itself test whether the conclusion is uniform as extremality is approached. We therefore performed a second scan at fixed $y = \omega/\kappa = (0.025, 0.05, 0.1)$ for $a/M = (0.98, 0.99, 0.995, 0.997, 0.999)$. Equation (92) predicts

$$M^{2n-2} \chi^{(n)}_{H,00} \sim c_n y^{2n-2} (M\kappa)^{2n-2}. \tag{95}$$

At the smallest y, fits to the three highest spins give powers 2.00734 and 4.01128, with $R^2 = 0.99999988$ and 0.99999993. The results therefore resolve suppression by κ^2 and κ^4, respectively. These paths provide numerical evidence for monopole suppression, not a proof of a uniform asymptotic expansion at extremality. The fixed-spin matching argument and the fixed-ω/κ tests have distinct domains of validity.

The new source calculations reuse the same homogeneous solution for every n. At the lowest representative point $(a/M, M\omega, N) = (0.5, 5 \times 10^{-4}, 176)$, the last 16 normalized phase-peeled Chebyshev coefficients are at most 5.2×10^{-14}. Across 1800 independent Chebyshev–Gauss validation points, the original field-dependent radial residual in Eq. (56) has median 2.7×10^{-10}, 95th percentile 2.1×10^{-7}, and a localized maximum 1.6×10^{-5}. Over both response scans the largest two-resolution change is 2.5×10^{-6}, the largest linear-balance error is 7.7×10^{-13}, and the largest Wronskian error is 1.1×10^{-14}. At the smallest responses the separately reconstructed reflected interference is cancellation limited and is not used: the power-law test uses the direct horizon Green integral, its two-resolution convergence, and the independent homogeneous diagnostics just quoted.

C. Near-extremal principal-mode enhancement

We next hold the asymptotic incident amplitude and $y = p/\kappa \neq 0$ fixed while approaching extremal rotation, with $m > 0$. The frequency then approaches $\omega_* = m/(2M)$. This path probes concentration of the incident-normalized field in the Kerr throat. We define

$$\sigma = \frac{r_+ - r_-}{M}, \qquad \widehat{\chi}_n = M^{2n-2} \frac{T^{(1)}_n}{T^{(0)}}. \tag{96}$$

Since $M\kappa = \sigma/[4(1 + \sigma/2)]$, the leading powers in σ and $M\kappa$ agree. In the following connection formulas we set $M = 1$. The normalization is unit ingoing horizon amplitude for $R_{\rm in}$ and unit outgoing infinity amplitude for $R_{\rm up}$.

1. *The throat equation and its two branches*

Let $X = (r - r_+)/\sigma$. The exact relations $\Delta = \sigma^2 X(X + 1)$ and $(r_+^2 + a^2)p = \sigma y/2$ give

$$K = \sigma \left(\frac{y}{2} + 2r_+ \omega X + \sigma \omega X^2 \right). \tag{97}$$

At fixed X, y the leading radial equation is

$$\frac{d}{dX} \left[X(X + 1) \frac{df}{dX} \right] + \left[\frac{(mX + y/2)^2}{X(X + 1)} - \lambda_* \right] f = 0, \tag{98}$$

TABLE X. Accepted low-frequency-side data displayed in Fig. 4. Only points satisfying $|T^{(1)}+R^{(1)}|\le 10^{-8}$ are retained. The last three columns give the linear transmission, first-order transmission, and absolute first-order balance residual at the lowest displayed frequency.

ℓ	displayed $M\omega$ range	points	$T^{(0)}_{\rm low}$	$T^{(1)}_{\rm low}$	$\lvert\delta_1\rvert_{\rm low}$
0	$[0.02, 0.09]$	15	6.92348×10^{-3}	1.67885×10^{-3}	1.4×10^{-13}
1	$[0.12, 0.28]$	17	9.73095×10^{-4}	1.78519×10^{-4}	3.7×10^{-14}
2	$[0.28, 0.46]$	19	4.35440×10^{-4}	6.58814×10^{-5}	2.0×10^{-14}

TABLE XI. Convergence audit at the lowest accepted displayed point for each multipole. The reference uses $(N, q, r_{\rm m}/M) = (256, 256, 40)$. The maxima include the active $(240, 192, 40)$ setting and the one-choice variations $(224, 256, 40)$, $(256, 192, 40)$, $(256, 256, 32)$, and $(256, 256, 48)$. The last column is the largest absolute first-order balance residual in the scan.

ℓ	$M\omega$	$T^{(1)}_{\rm ref}$	$\max\lvert\Delta T^{(1)}/T^{(1)}\rvert$	$\max\lvert\Delta R^{(1)}/R^{(1)}\rvert$	$\max\lvert\delta_1\rvert$
0	0.02	1.67885×10^{-3}	1.8×10^{-8}	2.0×10^{-8}	3.0×10^{-12}
1	0.12	1.78519×10^{-4}	6.7×10^{-9}	7.3×10^{-9}	3.8×10^{-13}
2	0.28	6.58814×10^{-5}	9.2×10^{-9}	4.6×10^{-9}	2.8×10^{-13}

where $\lambda_* = A_{\ell m}(m/2) - 3m^2/4$ in our angular convention. In the overlap region the indicial equation gives

$$h(h-1)+m^2-\lambda_*=0,\qquad h=\frac{1}{2}+\sqrt{\frac{1}{4}+A_{\ell m}(m/2)-\frac{7m^2}{4}}. \tag{99}$$

The two branches are X^{h-1} and X^{-h}. The principal class has $h=1/2+i\beta$, $\beta>0$, so both branches have the same $X^{-1/2}$ amplitude envelope. The real-weight class has distinct real exponents. This classification follows from the radial equation [19].

The horizon-ingoing solution is

$$f_y(X) = X^{-iy/2}(1+X)^{i(y/2-m)} \times {}_2F_1(h-im, 1-h-im; 1-iy; -X). \tag{100}$$

Continuation to large X yields $f_y \sim c_+ X^{h-1} + c_- X^{-h}$, with

$$c_+ = \frac{\Gamma(1-iy)\Gamma(2h-1)}{\Gamma(h-im)\Gamma[h-i(y-m)]},\qquad c_- = \frac{\Gamma(1-iy)\Gamma(1-2h)}{\Gamma(1-h-im)\Gamma[1-h-i(y-m)]}. \tag{101}$$

2. *Connection to asymptotically flat radiation*

At extremality set $u = r-1$. The critical-frequency outer equation becomes

$$u^2R'' + 2uR' + [m^2(1+u/2)^2 - \lambda_*]R = 0. \tag{102}$$

Setting $\Psi = uR$ and $z = -imu$ gives the Whittaker equation

$$\frac{d^2\Psi}{dz^2} + \left[-\frac{1}{4} + \frac{im}{z} + \frac{1/4-(h-1/2)^2}{z^2}\right]\Psi = 0. \tag{103}$$

The solutions normalized to u^{h-1} and u^{-h} are

$$R_+(u) = (-im)^{-h}u^{-1}M_{im,h-1/2}(-imu),\qquad R_-(u) = (-im)^{h-1}u^{-1}M_{im,1/2-h}(-imu). \tag{104}$$

Their incoming amplitudes are

$$U_+ = \frac{\Gamma(2h)}{\Gamma(h-im)}(-im)^{-h-im},\qquad U_- = \frac{\Gamma(2-2h)}{\Gamma(1-h-im)}(-im)^{h-1-im}. \tag{105}$$

These coefficients share a unit-modulus phase determined by the additive constant in r_*. We use the principal branch of $\log(-im)$ consistently. The near hypergeometric and outer Whittaker representations implement the leading near-extremal analytic connection. MST supplies the full-frequency series and connection structure[21]; its renormalized angular momentum ν and the throat index h label the respective representations.

In the overlap region $u \simeq \sigma X$, matching gives

$$B^{\rm inc} \sim c_+U_+\sigma^{1-h} + c_-U_-\sigma^h = \sigma^{1/2}\mathcal{D}(\theta),\qquad \theta = \beta\ln\sigma,\qquad \mathcal{D}(\theta) = c_+U_+e^{-i\theta} + c_-U_-e^{i\theta}. \tag{106}$$

Thus fixed-incident normalization yields

$$|R^{(0)}_H|^2 \sim \frac{\sigma^{-1}}{|\mathcal{D}(\theta)|^2}. \tag{107}$$

The near-critical horizon flux carries an additional factor $p \propto \sigma$. Consequently the linear absorption is finite and log-phase dependent, with $T^{(0)} \sim y/[m|\mathcal{D}|^2]$ in $M=1$ units.

TABLE XII. Scalar-monopole test of the $|\Phi|^{2n}\Phi$ infrared hierarchy. The fixed-ω slope is the range over the five spins; the fixed-y slope uses $y = 0.025$ and the three highest spins. The coefficient error is the largest quadratic-extrapolation error over spin. The final column is the largest two-resolution response change over both scans.

n	c_n	predicted power	fixed-ω fitted power	fixed-y κ power	max. coefficient error	max. $\delta_{N\chi_H}$
2	0.0631722552	2	2.01009–2.01016	2.00734	2.8×10^{-6}	2.5×10^{-6}
3	0.0172436384	4	4.01552–4.01561	4.01128	1.3×10^{-6}	7.7×10^{-9}

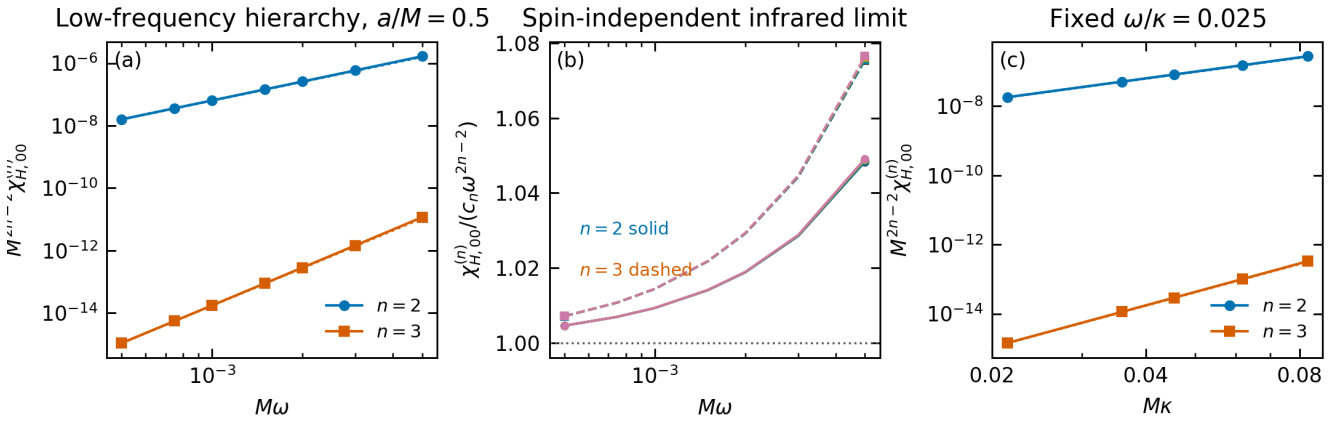


FIG. 3. Higher-power scalar-monopole response. (a) Direct horizon response for $n = 2, 3$ at $a/M = 0.5$; dotted lines are the analytic leading laws $c_n\omega^{2n-2}$. (b) The scaled response divided by c_n for five spins; solid circles and dashed squares denote $n = 2$ and 3. The near coincidence of the colored spin curves exhibits the common far-zone limit. (c) At fixed $\omega/\kappa = 0.025$, the response decreases toward extremality; dotted curves are $c_n(0.025\kappa)^{2n-2}$. Points show the $N = q = 176$ calculation; all pass the paired-resolution, linear-balance, and Wronskian tests described in the text.

3. *The nonlinear source integral and the envelope law*

Choose near solutions $\psi_+ \sim X^{h-1}$ and $\psi_- \sim X^{-h}$. The outgoing Whittaker solution $u^{-1}W_{im,h-1/2}(-imu)$ has overlap coefficients

$$g_+ = \frac{\Gamma(1-2h)}{\Gamma(1-h-im)}(-im)^h,$$
$$g_- = \frac{\Gamma(2h-1)}{\Gamma(h-im)}(-im)^{1-h}. \tag{108}$$

Its near-zone form is $R_{\rm up} \sim \sigma^{-1/2}\gamma_\theta$, where

$$\gamma_\theta = g_+e^{i\theta}\psi_+ + g_-e^{-i\theta}\psi_-. \tag{109}$$

The constant near Wronskian and its physical counterpart are

$$w_\theta = X(X+1)(f_y\gamma_\theta' - \gamma_\theta f_y'),$$
$$W \sim \sigma^{1/2}w_\theta. \tag{110}$$

The covariant source weight at the limiting horizon is $C_H^{(n)} = C_{\ell\ell m}^{(0,n)} + C_{\ell\ell m}^{(2,n)}$. Since $dr = \sigma\, dX$, the Green projection gives the finite leading unit-horizon coefficient

$$\mathcal{A}_n(\theta) = -\frac{C_H^{(n)}}{w_\theta}\int_0^\infty \gamma_\theta |f_y|^{2n} f_y\, dX. \tag{111}$$

At $X = 0$ the amplitude envelope is bounded. At large X the integrand envelope is X^{-n-1}, so the integral converges for every positive integer n. The factors $\sigma^{-1/2}$ from $R_{\rm up}$, σ from dr, and $\sigma^{-1/2}$ from $1/W$ cancel. The remaining real power in the fixed-incident response therefore comes from $|B^{\rm inc}|^{-2n}$:

$$\widehat{\chi}_n = \sigma^{-n}\mathcal{F}_n(y,\theta) + o(\sigma^{-n}),$$
$$\mathcal{F}_n(y,\theta) = \frac{2\,\mathrm{Re}\,\mathcal{A}_n(\theta)}{|\mathcal{D}(\theta)|^{2n}}. \tag{112}$$

The expansion applies on compact parameter sets where the matching denominators stay bounded away from zero. At a zero of $\mathrm{Re}\,\mathcal{A}_n$, the next term controls the flux response. Transforming $\theta \mapsto \theta + \pi$ changes the signs of γ, w and $\mathcal{D}$ while preserving $\mathcal{A}_n$ and $|\mathcal{D}|$. The leading coefficient hence has logarithmic period π/β. The envelope power is $-n$, and the two-branch interference determines its log-periodic modulation.

For reproducible coefficient evaluation, write $\gamma = a_\theta f_y + b_\theta f_y^*$. The horizon current gives $|c_+|^2 - |c_-|^2 = -y/(2\beta)$ and $w_\theta = iyb_\theta$. Let $G_+ = g_+e^{i\theta}$, $G_- = g_-e^{-i\theta}$

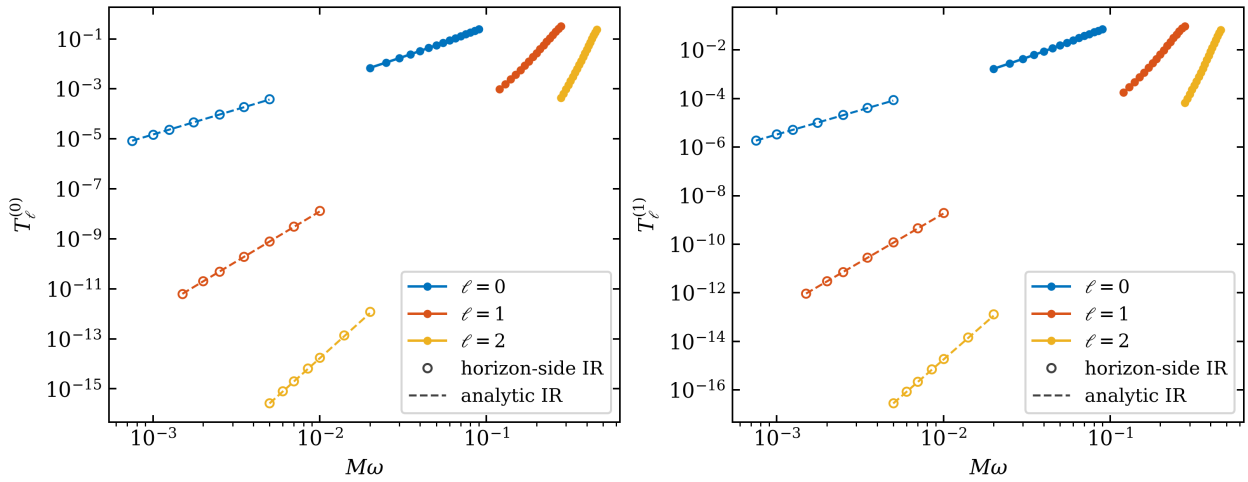


FIG. 4. Low-frequency Kerr diagnostics for $a = 0.5$, $m = 0$, and $\ell = 0, 1, 2$. Left: linear transmission coefficients. Right: self-channel nonlinear transmission coefficients. Filled circles and solid segments are the dense globally balanced sweep satisfying $|T^{(1)} + R^{(1)}| \le 10^{-8}$. Open circles are the higher-resolution member of the paired horizon-side infrared audit; dashed segments are the analytic leading laws $\mathcal{A}_\ell^{(i)}(M\omega)^{2\ell+2}$. The gap between the two sets marks loss of reflected-channel cancellation accuracy, not loss of convergence of the transmitted coefficient.

and $d_c = |c_+|^2 - |c_-|^2$. Then

$$
\begin{aligned}
a_\theta &= (G_+ c_+^* - G_- c_-^*)/d_c, \\
b_\theta &= (-G_+ c_- + G_- c_+)/d_c, \\
\mathcal{A}_n &= \frac{iC_H^{(n)}}{y}\left(\frac{a_\theta}{b_\theta} I_n^{\rm osc} + I_n^{\rm abs}\right), \\
I_n^{\rm osc} &= \int_0^\infty f_y^2 |f_y|^{2n}\, dX, \qquad I_n^{\rm abs} = \int_0^\infty |f_y|^{2n+2}\, dX.
\end{aligned}
\tag{113}
$$

Equations (101)–(113) specify the leading prediction with zero fitted parameters.

4. *Real weights and the perturbative domain*

For a real nondegenerate $h > 1/2$, the dominant incident connection has $B^{\rm inc} \sim c_+ U_+ \sigma^{1-h}$ and the incident-normalized near field scales as σ^{h-1}. The projected Green integrand in the overlap region scales as $X^{2n(h-1)-1}$. For $1/2 < h < 1$ the upper integral converges and the throat contribution has power $\sigma^{2n(h-1)}$. For $h > 1$, the matching upper limit $X \sim \sigma^{-1}$ supplies $\sigma^{-2n(h-1)}$; the resulting order-unity contribution is determined by the full outer matching. At $h = 1$ the upper integral has logarithmic behavior. This convergence criterion specifies when a local throat power controls the complete observable. The real-weight channel $(1, 1)$ has $h = 1.1706059743$ and belongs to the upper-limit-sensitive class.

Define $g_n = \epsilon_n/M^{2n-2}$ and the dimensionless complex horizon response $\widehat{\rho}_n$, with $\widehat{\chi}_n = 2\,\mathrm{Re}\,\widehat{\rho}_n$. The leading field correction has $\widehat{\rho}_n \sim \sigma^{-n}\mathcal{A}_n/|\mathcal{D}|^{2n}$. The relative horizon-field and flux corrections obey, respectively,

$$
\begin{aligned}
|g_n|\,|\mathcal{A}_{\rm in}|^{2n}|\widehat{\rho}_n| &\ll 1, \\
|g_n|\,|\mathcal{A}_{\rm in}|^{2n}|\widehat{\chi}_n| &\ll 1.
\end{aligned}
\tag{114}
$$

The field condition controls both the horizon amplitude and its phase. On principal-mode intervals with a bounded nonzero typical coefficient, the expansion parameter scales as $|g_n||\mathcal{A}_{\rm in}|^{2n}(M\kappa)^{-n}$. This identifies the shrinking weak-coupling domain near extremality.

TABLE XIII. Critical scalar angular eigenvalues and principal weights at $a/M = 1$, $M\omega = m/2$, with $h = 1/2 + i\beta$.

(ℓ, m)	$A_{\ell m}$	β
$(2, 2)$	5.8551625746	0.9459584692
$(3, 3)$	11.7447939547	1.9378354020
$(4, 4)$	19.6267345276	2.8501342902

5. *Expanded numerical verification*

The refined 81-point grid has a maximum paired real-response change of 6.55×10^{-9} and a complex-response change of 3.25×10^{-9}. Six representative cases also vary the panel order and lower integration limit at fixed $N = 448$, giving a maximum response change of 3.91×10^{-13}, and vary $r_{\rm match}/M = 8, 12, 16$, giving a maximum response range of 5.42×10^{-9}. The homogeneous fields are unchanged by the quadrature refinement; their maximum off-grid radial relative residual is 1.37×10^{-8} and their maximum coefficient tail is 6.82×10^{-14}. The maximum normalized interference-balance error is 2.47×10^{-9}.

TABLE XIV. Parameter-free leading matching at $a/M = 0.999999$. Relative matching differences are in percent. $\delta_{\rm pair}$ is the maximum change across all spins and orders under $(N, q) = (384, 32) \to (448, 48)$ with log-horizon quadrature. $\delta_{\rm old}$ compares the refined response with the previous global $q = 3072$ rule. Independent panel-order, lower-limit and matching-radius checks are described in the text.

(ℓ, m)	y	$n = 1$ (%)	$n = 2$ (%)	$n = 3$ (%)	$\delta_{\rm pair}$	$\delta_{\rm old}$
$(2, 2)$	-0.5	1.925	2.403	2.422	4.13×10^{-9}	6.49×10^{-8}
$(2, 2)$	0.5	0.825	0.964	0.894	8.43×10^{-11}	1.39×10^{-6}
$(2, 2)$	1	0.782	0.717	0.581	1.26×10^{-11}	4.15×10^{-6}
$(3, 3)$	-0.5	0.202	0.516	0.834	6.55×10^{-9}	6.16×10^{-8}
$(3, 3)$	0.5	0.111	0.307	0.506	1.26×10^{-10}	1.09×10^{-6}
$(3, 3)$	1	0.054	0.179	0.305	1.41×10^{-11}	1.26×10^{-5}
$(4, 4)$	-0.5	0.042	0.343	0.665	1.65×10^{-9}	1.17×10^{-7}
$(4, 4)$	0.5	0.066	0.294	0.537	9.19×10^{-10}	7.49×10^{-7}
$(4, 4)$	1	0.028	0.183	0.380	9.10×10^{-11}	2.27×10^{-5}

The original absolute flux gate is met by 60 points and is recorded separately from the local and paired-response criteria. The maximum absolute interference-balance error is 3.32×10^{-6}. The two analytic integral settings differ by at most 1.96×10^{-12} in the recorded relative integral diagnostic. The analytic log-period test has a maximum relative difference of 4.38×10^{-13}.

We evaluate the full covariant source for $(\ell, m) = (2, 2), (3, 3), (4, 4)$, $n = 1, 2, 3$, $y = -0.5, 0.5, 1$, and $a/M = 0.9999, 0.99999, 0.999999$. The grid has 81 physical parameter points. Refined production uses $N = 448$ and 48 Gauss nodes per logarithmic panel, with $r_{\rm match} = 12M$. A paired $N = 384$, 32-node calculation supplies an additional 162 solver calls across the grid; the previous 324 global-rule calls remain archived for comparison. For the inner Green integral we use $t = \ln[(r-r_+)/(r_+-r_-)]$, with panel lengths at most unity. The interval below $\delta = (r_+-r_-)e^{t_{\rm min}}$ is integrated using the leading horizon power. For example, with the same tortoise convention,

$$\int_0^\delta e^{-2ipr_*(r_++s)}\,ds = \frac{\delta e^{-2ipr_*(r_++\delta)}}{1-iy} + \mathcal{O}\left(\frac{\delta^2}{r_+-r_-}\right). \tag{115}$$

The regular prefactors are evaluated at the horizon in this short interval. The refined grid uses $t_{\rm min} = -34$; the representative checks vary $t_{\rm min} = -26, -30, -34$ and panel orders $32, 48, 64$. This procedure preserves the exact horizon in both the spectral equation and the Green integral. Stable evaluation in the positive distance s avoids subtracting nearly equal radii in the tortoise logarithms. The near integrals use $t = \ln X$ with two independent settings: $t \in [-30, 30]$, 12 Gauss nodes per unit interval, and $t \in [-36, 36]$, 20 nodes per interval. Special functions are evaluated with 32 decimal digits and accumulated quadratures use double precision. Angular eigenvalues and source projections use the critical spheroidal functions. This procedure fixes all prediction coefficients from the field equation and normalization.

The acceptance criteria for this large-response experiment combine paired changes of the real and complex horizon response, spectral tails, field-dependent off-grid radial residuals and separately recorded flux interference errors. Both absolute $|T_n^{(1)} + R_n^{(1)}|$ and relative $|T_n^{(1)} + R_n^{(1)}|/(|T_n^{(1)}| + |R_n^{(1)}|)$ are retained. The axisymmetric figures continue to use their original absolute gate. The principal-response figure uses the local and paired-response criteria reported with Table XIV; the CSV also identifies which points pass the original absolute gate. The thresholds are 10^{-5} for both paired horizon-response changes, 10^{-4} for the independent quadrature change, 10^{-6} for the off-grid radial residual, 10^{-8} for the spectral tail, and 10^{-6} for the normalized interference balance. The original absolute gate is 10^{-8}. These tolerances are applied uniformly across the grid. The physical flux correction multiplies each coefficient and its absolute numerical error by $\epsilon_n|\mathcal{A}_{\rm in}|^{2n}$.

The observed resolution changes quantify numerical error, while the difference from the parameter-free prediction quantifies the finite-σ correction to the leading matched approximation. The envelope in Eq. (112) follows analytically from the equal real parts of the two principal branches and the convergence of the nonlinear source integral. Three-spin slope estimates provide a finite-window diagnostic; the coefficient comparison supplies the direct test of the derived response function.

6. *Flux-response zeros and an enhanced phase response*

The complex coefficient provides an additional observable distinction. Define $\psi_n = \mathrm{Im}\,\widehat{\rho}_n$; its first-order transmitted phase shift is $g_n|\mathcal{A}_{\rm in}|^{2n}\psi_n$. Generalizing Eq. (60) to the self channel of $|\Phi|^{2n}\Phi$, replace P by $P_n = r^2C^{(0,n)} + a^2C^{(2,n)}$ and $|R^{\rm in}|^2$ by $|R^{\rm in}|^{2n}$ in $J_{\rm d}, J_{\rm o}$.

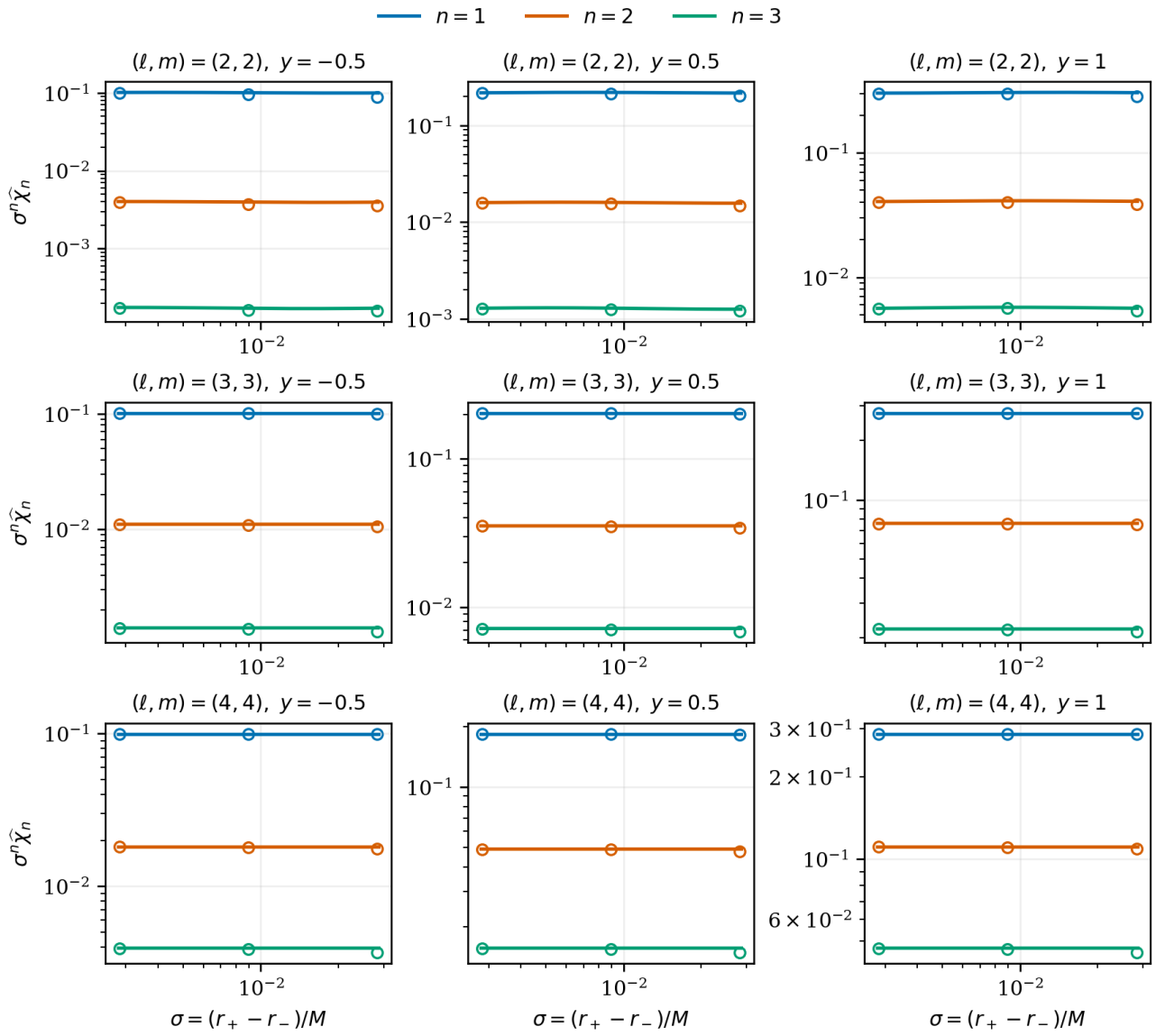


FIG. 5. Principal-mode response coefficients after removing the predicted envelope, $\sigma^n\widehat{\chi}_n$. Rows are $(2,2),(3,3),(4,4)$ and columns are $y=-0.5,0.5,1$. Colors label $n=1,2,3$. Lines are the parameter-free analytic functions evaluated at 180 display locations; symbols are the independently computed full Kerr spectral points. Each numerical series has three spins and paired spectral/logarithmic-panel resolutions. The solid functions retain the two-branch logarithmic phase. The comparison tests the leading coefficient as well as the envelope power.

Writing $q_{\rm r}=B^{\rm ref}/B^{\rm inc}$ gives

$$\widehat{\chi}_n=-\frac{M^{2n-2}\,\mathrm{Im}(q_{\rm r}J_{{\rm o},n})}{\omega|B^{\rm inc}|^{2n}},$$
$$\psi_n=\frac{M^{2n-2}[J_{{\rm d},n}+\mathrm{Re}(q_{\rm r}J_{{\rm o},n})]}{2\omega|B^{\rm inc}|^{2n}}. \tag{116}$$

Thus a zero of the imaginary interference projection makes the first-order energy-flux correction vanish while the real projection can retain a phase response. For principal modes both components carry the σ^{-n} envelope, with distinct phase-dependent coefficients.

We compute 27 further points for $(\ell,m)=(2,2)$, $n=1,2,3$, $y=1,2,4$, and the same three spins, each at paired resolutions. The maximum relative changes are 2.36×10^{-11} for the phase and 5.28×10^{-8} for the flux response. At $a/M=0.999999$, $n=3$, $y=2$, we find $\psi_3=2.4132\times10^4$ and $\widehat{\chi}_3=363.07$; at $y=4$ they are 1571.09 and -0.0102804, respectively. These samples establish phase-dominated response coefficients within the tested detuning range.

A resolved zero occurs near $y_\star=2.41954$ for this $n=3$ channel. Independent calculations at $N=448,576$, panel orders $48,64$, and $r_{\rm match}/M=8,12,16$ preserve the

opposite signs at $y_\star \pm 2\times10^{-4}$. At the common central estimate they give $|\widehat{\chi}_3| \le 5.73\times10^{-9}$ and $\psi_3 \simeq 1.0316\times10^4$. The sign bracket, rather than a relative error at zero, establishes the crossing. The first-order balance then gives $R_3^{(1)} = -T_3^{(1)}$, so both energy-flux derivatives vanish at the limiting zero. The phase coefficient retains its finite value there.

We independently reconstruct the sourced field on $1 \le X \le 8$ at this detuning and differentiate its Chebyshev interpolant. At reconstruction degrees $16, 24, 32, 48, 64$, the maximum field-dependent relative residuals are 0.128, 9.23×10^{-4}, 4.77×10^{-6}, 9.10×10^{-9}, and 6.62×10^{-8}, respectively. The coefficient tail falls from 3.84×10^{-3} to 4.33×10^{-15}; the full degree sequence displays the high-degree differentiation floor. This local test includes the source term in the residual denominator and supplements the global integral and sign-bracket checks. The horizon perturbative condition at a flux zero continues to involve $|g_n||\mathcal{A}_{\rm in}|^{2n}|\widehat{\rho}_n|$. Finite-coupling energy-flux changes require the second-order field as well as the square of the first-order field.

D. Kerr-to-Schwarzschild continuity

As a limiting-case check independent of external reference calculations, we hold $(\ell, m, M\omega) = (2, 0, 0.52)$ fixed and sweep the dimensionless spin from $a/M = 0$ to 0.5. The scalar spheroidal eigenvalue, the covariant source projection, the scattering amplitudes, and both flux coefficients are recomputed at every spin. At $a = 0$ the angular basis becomes spherical and the endpoint formulation reduces to the Schwarzschild spectral route; no radial endpoint is removed or replaced by a finite horizon cutoff. Figure 7 shows that $T^{(1)}$ and $-R^{(1)}$ vary smoothly from the $a = 0$ value, while Table XV records the balance and Wronskian diagnostics for the same points. The largest absolute first-order balance residual in this scan is 1.4×10^{-12} and the largest linear balance residual is below 1.2×10^{-11}. This experiment is a continuity and implementation check; it is not used as an external reference result or as a claim that the Kerr and Schwarzschild coefficient conventions are identical without the conversion in Appendix A.

V. DISCUSSION AND CONCLUSIONS

The complex fixed-incident response connects the frequency profile of nonlinear Kerr scattering to the wave amplitude available to drive the local interaction. Resonance peaks and post-resonance decay describe its finite-frequency structure. The fixed-spin infrared limit determines how the source scales after the penetration factor cancels in the fractional response. The fixed-p/κ near-extremal limit determines how throat concentration enhances that response. The Green-function projection and its flux–phase decomposition provide the common framework for these three regimes. This organization follows the Schwarzschild scattering construction while retaining the Kerr horizon frequency, spheroidal functions, and covariant source. All coefficients use the same fixed-incident normalization.

The cubic axisymmetric sweep at $a/M = 0.5$ resolves peak centers at $M\omega_p = 0.1306, 0.3101, 0.5026$ for $\ell = 0, 1, 2$. The centers vary by at most 6.7×10^{-4} under the stated fit-window changes and lie above the corresponding fundamental scalar QNM real parts by $16.2\%, 4.1\%, 2.2\%$. The nonlinear source projection and its phase enter the same Green coefficient as the homogeneous scattering kernel. The resulting source-weighted resonance provides a real-frequency nonlinear observable on the QNM scale. The fitted widths describe the measured profile; extracting modal residues requires a complex-frequency Green-function projection.

The post-resonance tails favor an exponential over a power law, with $\Delta\mathrm{AICc} = 97.9, 48.2, 23.8$ in the resolved windows. The exact decomposition in Eq. (60) attributes the first-order flux correction to the linear reflected amplitude times an oscillatory nonlinear source projection. It also separates a direct phase-shift contribution. The Schwarzschild-inspired $(M\omega)^{-1/2}$ prefactor improves the tail residuals for $\ell = 1, 2$ and changes the fitted decay scales in all three channels. For $\ell = 0$, its fitted temperature is within 1.3% of the Kerr T_H; for $\ell = 1, 2$, a free scale captures the residual modal dependence. The numerical conclusion is exponential suppression in the stated post-resonance windows, with a quantitatively tested dependence on the algebraic prefactor. A Kerr asymptotic exponent requires the corresponding complex-radial analysis of the reflection amplitude and source integral.

At fixed subextremal spin, the infrared response has an interaction-order hierarchy. At fixed nonzero p/κ, principal-mode scattering near extremality has an inverse-surface-gravity envelope with logarithmic modulation. The matched source integrals and asymptotically flat incident normalization determine both limits.

The flux formulation separates three quantities that can otherwise be confused: a unit-horizon Green coefficient, the relative fixed-incident field response, and the physical correction multiplied by the coupling and incident amplitude. The first quantity may grow at low frequency without producing a divergent fractional absorption correction. The exact identities $T^{(0)} + R^{(0)} = 1$ and $T^{(1)} + R^{(1)} = 0$ express energy conservation; they provide global checks alongside independent local residual, spectral-tail, and quadrature-refinement diagnostics.

For the cubic axisymmetric self channels, matched asymptotics give $T^{(0)}, T^{(1)} \propto (M\omega)^{2\ell+2}$ at fixed subextremal spin. Rotation changes the absolute coefficients, whereas the limiting ratio is fixed by the far-zone Bessel integral. In the monopole, $\rho_{H,00} \to \ln 2/(2\pi) + i/4$ and $\chi_{H,00} \to \ln 2/\pi$. The real part governs the absorption change and the imaginary part the horizon phase change. This distinguishes the nonlinear response from a state-

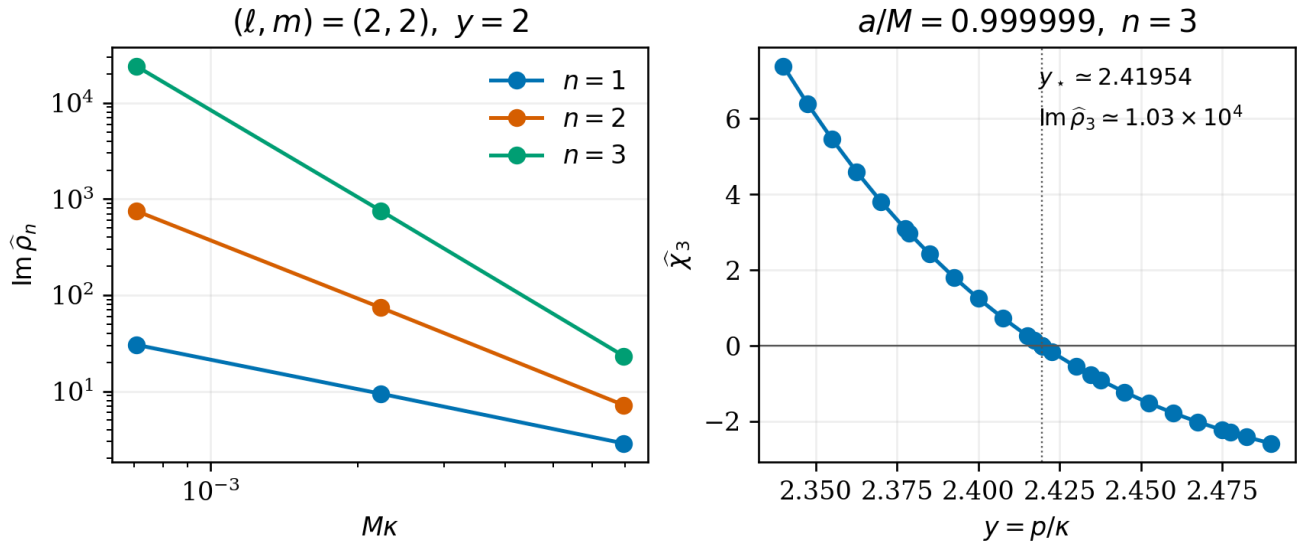


FIG. 6. Phase and flux components of the nonlinear response. Left: transmitted phase coefficients for $(2, 2)$ at fixed $y = 2$ and $n = 1, 2, 3$ across three spins. Right: the $n = 3$ flux-response crossing at $a/M = 0.999999$; circles include 21 uniformly spaced direct evaluations on $2.34 \le y \le 2.49$ and the root-search samples. Joining segments are visual guides. The phase coefficient at the crossing is about 1.03×10^4. Separate degree, panel-order and matching-radius checks bracket the zero as described in the text.

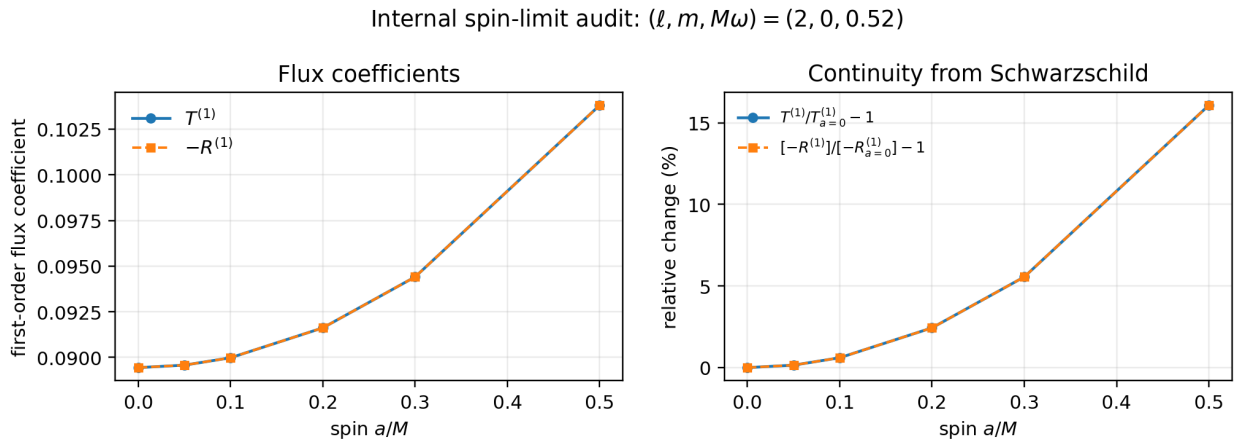


FIG. 7. Internal Kerr-to-Schwarzschild continuity test at fixed $(\ell, m, M\omega) = (2, 0, 0.52)$. The first-order transmission coefficient and $-R^{(1)}$ remain paired by the flux balance while changing smoothly with a/M. The right panel normalizes both coefficients to the $a = 0$ value; all points use the endpoint-inclusive spectral solver and are accepted by the first-order balance check.

ment about transmission alone.

The general monopole interaction gives $M^{2n-2}\chi^{(n)}_{H,00} \sim c_n(M\omega)^{2n-2}$ with $c_n > 0$. The distinction between $n = 1$ and $n > 1$ is an observable infrared hierarchy: the cubic fractional response remains finite, while the higher-power responses vanish. The $n = 2, 3$ numerical experiments evaluate the full source independently of the asymptotic law. The fitted powers characterize the finite frequency windows. At leading order in this fixed-spin infrared limit, rotation changes the absolute penetration amplitude and the far-zone integral fixes the fractional-response coefficient independently of spin.

For principal modes, $h = 1/2 + i\beta$ gives equal real powers for both overlap branches. Their matching produces $|B^{\mathrm{inc}}|^{-2} \sim \sigma^{-1}|\mathcal{D}|^{-2}$. The unit-horizon nonlinear Green integral converges for every $n \ge 1$, so the fixed-incident response has $\widehat{\chi}_n = \sigma^{-n}\mathcal{F}_n(y, \beta \ln \sigma) + o(\sigma^{-n})$. The real part of h fixes the envelope; its imaginary part and the two connection coefficients fix the log-periodic modulation. The derived Γ-function connections and source integrals determine the coefficient with zero fitted parameters. Fresh calculations for $(2, 2), (3, 3), (4, 4)$, $n = 1, 2, 3$, and $y = -0.5, 0.5, 1$ test that prediction across three spins. At $a/M = 0.999999$, the leading matching differences range from about 0.03% to 2.42%, whereas independent numerical refinements have a max-

TABLE XV. Internal Kerr-to-Schwarzschild spin-limit audit for $(\ell, m, M\omega) = (2, 0, 0.52)$. The entries use the same fixed-incident normalization as the production coefficients. The two balance columns are $T^{(0)} + R^{(0)} - 1$ and $T^{(1)} + R^{(1)}$, respectively.

a/M	$T^{(1)}$	$R^{(1)}$	$\lvert\delta_0\rvert$	$\lvert\delta_1\rvert$	$\lvert\delta_W\rvert$
0.00	8.94395×10^{-2}	-8.94395×10^{-2}	1.1×10^{-12}	3.8×10^{-13}	2.6×10^{-16}
0.05	8.95742×10^{-2}	-8.95742×10^{-2}	9.8×10^{-12}	1.1×10^{-12}	3.3×10^{-16}
0.10	8.99794×10^{-2}	-8.99794×10^{-2}	3.7×10^{-12}	4.9×10^{-13}	0
0.20	9.16178×10^{-2}	-9.16178×10^{-2}	4.3×10^{-13}	7.6×10^{-14}	1.8×10^{-16}
0.30	9.44095×10^{-2}	-9.44095×10^{-2}	1.1×10^{-11}	1.3×10^{-12}	1.8×10^{-16}
0.50	1.03814×10^{-1}	-1.03814×10^{-1}	6.0×10^{-12}	1.2×10^{-12}	0

imum paired response change of 6.55×10^{-9}. Six representative cases have logarithmic-panel and lower-limit changes below 3.92×10^{-13}; their matching-radius response range is below 5.42×10^{-9}. These comparisons resolve finite-σ matching corrections separately from numerical discretization effects.

The real-weight source convergence criterion supplies the corresponding scope: $1/2 < h < 1$ gives a convergent throat power, $h = 1$ gives a logarithmic upper integral, and $h > 1$ requires the outer-region contribution at leading order. The principal envelope thus has a precise modal origin. It also identifies a shrinking perturbative domain, $|g_n||\mathcal{A}_{\rm in}|^{2n}(M\kappa)^{-n} \ll 1$, on intervals with a bounded nonzero response coefficient.

The broader detuning study resolves a complementary phase-dominated regime. At the $n = 3$, $(2, 2)$ crossing near $p/\kappa = 2.41954$, the first-order horizon and reflected energy-flux derivatives vanish and the transmitted phase coefficient is about 1.03×10^4. Equation (116) identifies the crossing with the imaginary interference projection and retains the real phase projection. This response zero is independently bracketed under degree, quadrature and matching-radius variation and checked with a local sourced residual in the throat. It provides a concrete example in which the complex field response controls the weak-coupling domain at a vanishing first-order energy-flux response. Computing the finite-coupling flux near that detuning requires a second-order or nonlinear calculation.

The incident-power condition is essential to the near-extremal envelope. An excitation of duration $\tau \sim \kappa^{-1}$ with fixed incident energy has a different amplitude scaling from a fixed-power drive. Determining its nonlinear response requires the frequency convolutions of a wave-packet source. This is a separate extension of the stationary results, alongside the dynamical near-extremal problems in Refs. [15–17].

The cubic absorption correction is positive over the retained axisymmetric samples in our signed-coupling convention. A conventional positive quartic potential corresponds to the opposite sign of the equation-level coupling and reverses this physical correction. The first-order reflected-flux correction follows from $T^{(1)} + R^{(1)} = 0$. This connects each resonance and tail feature directly to energy redistributed between the horizon and infinity.

The numerical error budget is resolved by observable and parameter window. Near the finite-frequency peak, off-grid residuals, coefficient tails, and paired resolutions test the radial solution. In the exponential-tail windows, inner Green quadrature limits the response accuracy. At the deepest infrared points, reflected interference suffers cancellation, so the horizon response is tested separately. The localized residual maximum in the higher-power monopole diagnostic is 1.6×10^{-5}, illustrating the value of a local accuracy test alongside flux conservation and fit quality. The principal grid has a maximum homogeneous off-grid radial residual of 1.37×10^{-8} and a coefficient tail of 6.82×10^{-14}. A separate interior reconstruction test is reported in Appendix B; it tests the reconstructed first-order field on the stated interior interval alongside Green-quadrature refinement and boundary-flux checks.

The principal result extends the linear near-horizon connection framework [19] by including the covariant nonlinear source and its Green projection. The numerical study covers the stated self channels and fixed-detuning paths. Further tests include higher modal indices, additional detunings, the complete mode-coupled first-order field, and a time-domain assessment of the shrinking perturbative domain. A real scalar with a Φ^3 source additionally generates harmonics such as 3ω; the present single-frequency closure follows from the complex field and its $U(1)$-invariant interaction. Nonlinear saturation and gravitational backreaction require extensions beyond this first-order fixed-background calculation.

DATA AND CODE AVAILABILITY

The figures, numerical tables, and producing scripts are retained in the local review package described in the accompanying reproducibility record. The axisymmetric numerical inputs are unchanged by this editorial revision. A versioned public release and persistent identifier remain to be supplied before submission. Completed solver calls and accepted manuscript points are distinguished in the CSV data. The fresh principal grid includes its input hashes, analytic coefficient checks, and separate spectral and quadrature comparisons.

ACKNOWLEDGMENTS

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OpenAI Codex assisted with drafting, derivation cross-checks, code review, and numerical postprocessing. Responsibility for the scientific content, references, and conclusions rests with the human authors.

Appendix A: Flux normalizations

1. Projected Kerr source

The scalar wave equation separates only after multiplying by $\Sigma = r^2 + a^2\cos^2\theta$. At first order this gives

$$\Sigma\Box\Phi^{(1)} = -\Sigma|\Phi^{(0)}|^2\Phi^{(0)}. \tag{A1}$$

Using

$$\begin{aligned}\Phi^{(0)} &= e^{-\mathrm{i}\omega t+\mathrm{i}m\phi}S_{\ell m}R_{\ell m\omega},\\ \Phi^{(1)} &= e^{-\mathrm{i}\omega t+\mathrm{i}m\phi}\sum_{\ell'}S_{\ell' m}R^{(1)}_{\ell'|\ell m\omega},\end{aligned} \tag{A2}$$

and projecting with $S^*_{\ell' m}$ gives

$$\mathcal{L}_{\ell' m\omega}R^{(1)}_{\ell'|\ell m\omega} = -\left(r^2C^{(0)}_{\ell'\ell m}+a^2C^{(2)}_{\ell'\ell m}\right)|R^{\mathrm{in}}_{\ell m\omega}|^2R^{\mathrm{in}}_{\ell m\omega}. \tag{A3}$$

Here

$$C^{(j)}_{\ell'\ell m} = \int \mathrm{d}\Omega\,\cos^j\theta\,S^*_{\ell' m}|S_{\ell m}|^2S_{\ell m}, \qquad j=0,2. \tag{A4}$$

This is the point at which Kerr differs from the spherical Schwarzschild reduction: a single incident spheroidal channel sources a matrix of target channels with the same m. Only the target channel $\ell'=\ell$ contributes linearly to the total flux, because only that channel interferes with the zeroth-order outgoing field. Off-diagonal channels contribute to the energy flux first at order ϵ^2.

2. Linear current

The scalar radial current is

$$J[R] = \frac{\Delta}{2\mathrm{i}}\left(R^*\frac{\mathrm{d}R}{\mathrm{d}r}-R\frac{\mathrm{d}R^*}{\mathrm{d}r}\right). \tag{A5}$$

Using the asymptotics in Eq. (22), the far-zone current is

$$J_\infty = -\omega|B^{\mathrm{inc}}|^2+\omega|B^{\mathrm{ref}}|^2. \tag{A6}$$

At the horizon, Eq. (21) gives

$$J_H = -p(r_+^2+a^2). \tag{A7}$$

The equality $J_\infty = J_H$ gives Eq. (25).

3. Fixed-incident nonlinear flux

The Green-function amplitudes $A^{(1)}_{\mathrm{ref}}$ and $A^{(1)}_H$ are computed with the zeroth-order solution normalized to unit horizon amplitude. Scattering coefficients, however, are defined at fixed incident amplitude. Therefore we rescale the zeroth-order solution by

$$\alpha = \frac{1}{B^{\mathrm{inc}}}. \tag{A8}$$

Because the source is cubic, the first-order field scales as $|\alpha|^2\alpha$. The fixed-incident self-channel field has the asymptotic form

$$\begin{aligned}R_{\mathrm{fix}} &\sim \frac{e^{-\mathrm{i}\omega r_*}}{r}\\ &+\left(\frac{B^{\mathrm{ref}}}{B^{\mathrm{inc}}}+\epsilon\frac{A^{(1)}_{\mathrm{ref}}}{|B^{\mathrm{inc}}|^2B^{\mathrm{inc}}}\right)\frac{e^{+\mathrm{i}\omega r_*}}{r}, \qquad r\to\infty,\end{aligned} \tag{A9}$$

and

$$R_{\mathrm{fix}} \sim \left(\frac{1}{B^{\mathrm{inc}}}+\epsilon\frac{A^{(1)}_H}{|B^{\mathrm{inc}}|^2B^{\mathrm{inc}}}\right)e^{-\mathrm{i}pr_*}, \qquad r\to r_+. \tag{A10}$$

The transmitted flux coefficient is therefore

$$\begin{aligned}T &= \frac{p(r_+^2+a^2)}{\omega}\left|\frac{1}{B^{\mathrm{inc}}}+\epsilon\frac{A^{(1)}_H}{|B^{\mathrm{inc}}|^2B^{\mathrm{inc}}}\right|^2+\mathcal{O}(\epsilon^2)\\ &= T^{(0)}+\epsilon\frac{2p(r_+^2+a^2)}{\omega|B^{\mathrm{inc}}|^4}\,\mathrm{Re}\,A^{(1)}_H+\mathcal{O}(\epsilon^2),\end{aligned} \tag{A11}$$

which gives Eq. (46). Similarly,

$$\begin{aligned}R &= \left|\frac{B^{\mathrm{ref}}}{B^{\mathrm{inc}}}+\epsilon\frac{A^{(1)}_{\mathrm{ref}}}{|B^{\mathrm{inc}}|^2B^{\mathrm{inc}}}\right|^2+\mathcal{O}(\epsilon^2)\\ &= R^{(0)}+\epsilon\frac{2\,\mathrm{Re}(\overline{B^{\mathrm{ref}}}\,A^{(1)}_{\mathrm{ref}})}{|B^{\mathrm{inc}}|^4}+\mathcal{O}(\epsilon^2),\end{aligned} \tag{A12}$$

which gives Eq. (47). Conservation of the radial current for the inhomogeneous solution with no incoming first-order radiation then yields the first-order balance law Eq. (53). If the prescribed incident amplitude is $\mathcal{A}_{\mathrm{in}}$ rather than unity, both first-order boundary amplitudes acquire the factor $|\mathcal{A}_{\mathrm{in}}|^2\mathcal{A}_{\mathrm{in}}$ and division by the incident flux leaves the factor $|\mathcal{A}_{\mathrm{in}}|^2$ in Eq. (52).

4. Schwarzschild normalization correspondence

The reference manuscript[18] writes $\Box\phi = \lambda_{\mathrm{ref}}|\phi|^{2N-2}\phi$. Its interaction index and equation-level coupling are related to ours by

$$N = n+1, \qquad \epsilon_n = -\lambda_{\mathrm{ref}}. \tag{A13}$$

The amplitude relations below use a common source-sign convention. When applying them directly to a first-order

amplitude quoted per $\lambda_{\rm ref}$, the opposite source sign must also be included. The physical flux correction agrees only after both coupling and normalization have been translated.

The flux coefficients in this paper use unit incident amplitude. This point matters when comparing them with a reduced Schwarzschild calculation that quotes the first-order Green coefficient with the zeroth-order solution normalized at the horizon. In the nonrotating limit, write the scalar field in the reduced convention as $\Phi = e^{-i\omega t}Y_{\ell m}u/r$ and let $\widehat{I}$ and $\widehat{R}$ denote the corresponding incident and reflected amplitudes. The present unit-horizon Teukolsky variable has $R^{\rm in} = r_+u/r$ at zeroth order, so that

$$B^{\rm inc} = r_+\widehat{I}, \qquad B^{\rm ref} = r_+\widehat{R}. \tag{A14}$$

The cubic source is homogeneous of degree three. Consequently the horizon and infinity Green amplitudes obey $A_H^{(1)} = r_+^2\widehat{A}_H^{(1)}$ and $A_{\rm ref}^{(1)} = r_+^3\widehat{A}_{\rm ref}^{(1)}$ when the reduced solution is horizon normalized. For $a = 0$ (so $p = \omega$), Eqs. (46) and (47) therefore become

$$T^{(1)}_{\text{unit-inc}} = \frac{2\,\mathrm{Re}\,\widehat{A}_H^{(1)}}{|\widehat{I}|^4}, \tag{A15}$$

$$R^{(1)}_{\text{unit-inc}} = \frac{2\,\mathrm{Re}(\widehat{R}^*\,\widehat{A}_{\rm ref}^{(1)})}{|\widehat{I}|^4}. \tag{A16}$$

If the reduced Schwarzschild coefficients are instead defined with the common unit-horizon convention, $\widehat{T}^{(1)} = 2\,\mathrm{Re}\,\widehat{A}_H^{(1)}/|\widehat{I}|^2$ and analogously for $\widehat{R}^{(1)}$, then

$$T^{(1)}_{\text{unit-inc}} = T^{(0)}\widehat{T}^{(1)}, \qquad R^{(1)}_{\text{unit-inc}} = T^{(0)}\widehat{R}^{(1)}. \tag{A17}$$

Thus a direct comparison of numerical values of $T^{(1)}$ between the two conventions is meaningful only after this normalization conversion. The physical first-order flux correction and the balance law are unchanged.

Appendix B: Independent local and endpoint validation

The main text combines axisymmetric infrared and finite-frequency data with the freshly computed principal-mode grid in Sec. IV C. This appendix documents the endpoint identities and the representative local sourced-field tests that preceded the expanded principal-mode calculation.

1. Cancellation-free peeled coefficients

Write $H = r^2 + a^2$, $K = \omega H - am$, and $p = \omega - m\Omega_H$. If $R = Fu$, the coefficient multiplying u is

$$B_0 = \Delta(q' + q^2) + \Delta' q + K^2/\Delta - \lambda, \qquad q = F'/F. \tag{B1}$$

For $F = r^{-1}\exp(i\sigma\omega r_*)$, $\sigma = \pm 1$, direct substitution and cancellation before floating-point evaluation give

$$B_0^\sigma = \frac{am(am - 2\omega H)}{\Delta} - \lambda - \frac{2i\sigma\omega a^2}{r} - \frac{2M}{r} + \frac{2a^2}{r^2}. \tag{B2}$$

Consequently the exact infinity row is $B_0(0) = -\lambda - 2am\omega = -(A + a^2\omega^2)$, not $-\lambda$. The difference vanishes for $m = 0$ or $a = 0$. For $F = \exp(-ipr_*)$, the identity $K - pH = (\omega - p)(r - r_+)(r + r_+)$ gives

$$B_0^{\rm in} = \frac{(\omega - p)(r + r_+)(K + pH)}{r - r_-} - 2ipr - \lambda. \tag{B3}$$

This removes the apparent division by $r - r_+$ without moving the horizon boundary. The normalization remains $u(1) = 1$ and the degenerate ODE row remains in the matrix. Neither expression changes the differential equation.

We tested $a/M = 0.99, 0.999$, $\ell = m = 1, 2$, and $n = 1, 2$ at $p/\kappa = 0.5$. The spectral degrees were $128, 176, 224$, with $r_{\rm match}/M = 12$ and Gauss orders $2N$. A further run at $N = 224$ used the erroneous infinity row solely to quantify its effect. The corrected maximum homogeneous radial relative residual across these probes was 3.2×10^{-9}; the maximum peeled-amplitude relative residual was 5.2×10^{-8}. Both denominators contain the corresponding field terms. The erroneous row changes the tested responses only at the 10^{-8} level or below, but produces spurious outer spectral tails. The freshly computed grid in Sec. IV C uses the corrected endpoint coefficients throughout.

At $a/M = 0.999$, $\ell = m = 2$, the $N = 176 \to 224$ fractional change in the cubic response was 1.6×10^{-6} when the quadrature order also changed. At fixed $N = 224$, increasing the Gauss order from 448 to 1792 gives a similar change, identifying quadrature as a separate error source. Matching radii $8M, 12M, 16M$ were also checked independently. These tests quantify consistency at the stated finite parameters.

2. Local residual of the sourced field

Let $Q = (r^2C_0 + a^2C_2)|R_0|^{2n}R_0$ denote the projected source without its minus sign, and $V = K^2/\Delta - \lambda$. With unit-horizon $R_{\rm in}$ and unit-outgoing $R_{\rm up}$, reconstruct

$$R_1(r) = -\frac{R_{\rm up}(r)I_{\rm in}(r) + R_{\rm in}(r)I_{\rm up}(r)}{W}, \tag{B4}$$

where

$$I_{\rm in}(r) = \int_{r_+}^{r} R_{\rm in}(s)Q(s)\,ds, \tag{B5}$$

$$I_{\rm up}(r) = \int_{r}^{\infty} R_{\rm up}(s)Q(s)\,ds, \tag{B6}$$

$$W = \Delta(R_{\rm in}R'_{\rm up} - R_{\rm up}R'_{\rm in}). \tag{B7}$$

TABLE XVI. Independent sourced-field residual and coefficient tail on $[3M, 8M]$. D is the reconstruction degree; the homogeneous degree is fixed. The tail is the maximum of the last eight coefficient magnitudes divided by the maximum coefficient magnitude.

Case	D	$\max \delta_1$	median δ_1	tail
A	16	1.05×10^{-4}	4.11×10^{-6}	1.71×10^{-5}
A	24	1.52×10^{-7}	3.97×10^{-9}	5.78×10^{-9}
A	32	1.85×10^{-9}	1.31×10^{-11}	2.50×10^{-12}
A	48	4.73×10^{-9}	3.36×10^{-11}	4.42×10^{-15}
A	64	5.90×10^{-9}	5.51×10^{-11}	3.74×10^{-15}
B	16	9.87×10^{-5}	3.63×10^{-6}	1.93×10^{-5}
B	24	1.42×10^{-7}	3.50×10^{-9}	6.50×10^{-9}
B	32	2.25×10^{-9}	1.64×10^{-11}	2.81×10^{-12}
B	48	4.71×10^{-9}	3.69×10^{-11}	6.51×10^{-15}
B	64	6.38×10^{-9}	6.47×10^{-11}	5.35×10^{-15}
C	16	4.97×10^{-5}	5.87×10^{-7}	3.56×10^{-4}
C	24	9.77×10^{-11}	2.37×10^{-12}	2.54×10^{-8}
C	32	3.23×10^{-10}	2.22×10^{-12}	2.38×10^{-14}
C	48	1.69×10^{-9}	6.69×10^{-12}	2.24×10^{-14}
C	64	1.16×10^{-9}	1.29×10^{-11}	2.70×10^{-14}

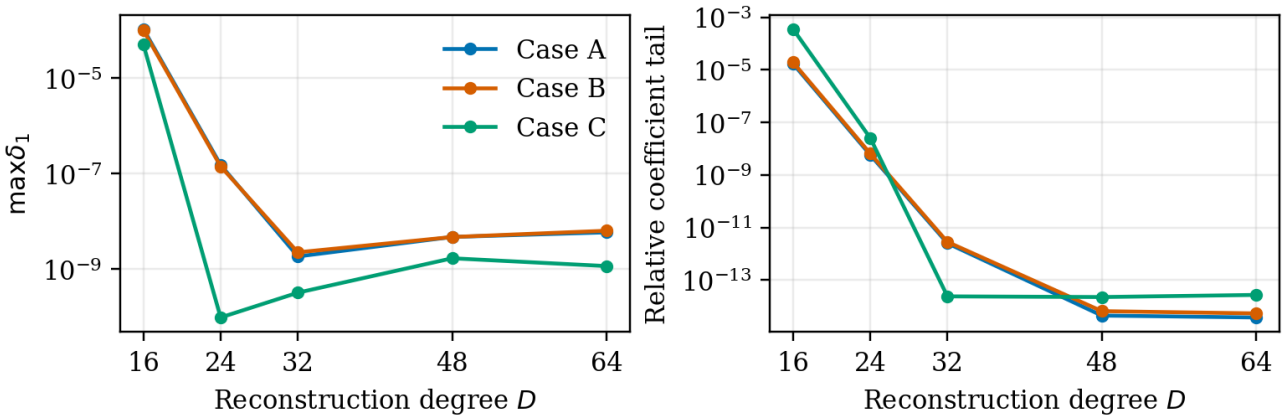


FIG. 8. Independent local first-order-field validation. Left: maximum field-dependent sourced residual on 600 validation nodes. Right: relative coefficient tail. Cases and degrees are identical to Table XVI. The high-degree residual floor is shown explicitly; a smaller coefficient tail does not guarantee a smaller second-derivative residual.

To avoid replacing a numerical test by a Green-function identity, we evaluate these integrals at reconstruction nodes and differentiate their Chebyshev interpolant, rather than setting R_1'' from the ODE. The independent local diagnostic is

$$\delta_1(r) = \frac{|\Delta R_1'' + \Delta' R_1' + V R_1 + Q|}{|\Delta R_1''| + |\Delta' R_1'| + |V R_1| + |Q|}. \tag{B8}$$

The field factors in this denominator are essential for invariance under a common rescaling of the first-order field and its source.

The homogeneous degree is 224. The initial horizon-to-$3M$ integrals use 1024 open Gauss nodes, the local cumulative increments use 128 nodes, and the global outgoing-source integral retains the infinite-tail calculation. We reconstruct on $3M \le r \le 8M$ with degrees $16, 24, 32, 48, 64$ and validate at 600 Chebyshev–Gauss points different from the reconstruction nodes. Cases A and B have $a/M = 0.5$, $\ell = m = 0$, $M\omega = 0.1$, with $n = 1, 2$, respectively. Case C has $a/M = 0.99$, $\ell = m = 2$, $p/\kappa = 0.5$, and $n = 1$. Table XVI reports all reconstruction degrees; none was selected by minimizing the residual after the calculation.

The decrease at low degree followed by a nonmonotonic floor is consistent with spectral truncation giving way to numerical interpolation and differentiation error. At degree 32, maxima are below 2.3×10^{-9} for all three cases. This verifies the local sourced equation on the stated interval, not the accuracy of the accumulated homogeneous integration constants or the omitted asymptotic inter-

vals. Global flux/quadrature checks are still needed. No global machine-precision accuracy claim follows from this test.

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